

Installing MantaRay HW Manager

DN1000058770

Issue 20

Approved on 2026-04-10

MantaRay HW Manager

Rel. 26R2-NM

Operating Documentation, Issue 01

Nokia is committed to diversity and inclusion. We are continuously reviewing our customer documentation and consulting with standards bodies to ensure that terminology is inclusive and aligned with the industry. Our future customer documentation will be updated accordingly.

This document includes Nokia proprietary and confidential information, which may not be distributed or disclosed to any third parties without the prior written consent of Nokia. This document is intended for use by Nokia's customers ("You"/"Your") in connection with a product purchased or licensed from any company within Nokia Group of Companies. Use this document as agreed. You agree to notify Nokia of any errors you may find in this document; however, should you elect to use this document for any purpose(s) for which it is not intended, You understand and warrant that any determinations You may make or actions You may take will be based upon Your independent judgment and analysis of the content of this document.

Nokia reserves the right to make changes to this document without notice. At all times, the controlling version is the one available on Nokia's site.

No part of this document may be modified.

AI Usage Disclaimer

The content of this documentation is provided exclusively for the intended Customer use of related Nokia product or service and is subject to the Terms of Use of the Nokia Support Portal.

No Use with Artificial Intelligence (AI):

Unless explicitly authorized in writing by Nokia, the Customer shall not use any part of this documentation—including but not limited to text, data, images, or other materials—for any purpose involving Artificial Intelligence (AI), including but not limited to:

- Training AI models or algorithms
- Processing, analyzing, or generating content using AI
- Extracting, scraping, or copying content using AI
- Creating derivative works or adaptations using AI
- Simulating user interaction to access, retrieve, or download content beyond standard portal features ("screen scraping")
- Integrating or using this content with AI-powered tools, including chatbots, virtual assistants, or generative AI systems

NO WARRANTY OF ANY KIND, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY OF AVAILABILITY, ACCURACY, RELIABILITY, TITLE, NON-INFRINGEMENT, MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, IS MADE IN RELATION TO THE CONTENT OF THIS DOCUMENT. IN NO EVENT WILL NOKIA BE LIABLE FOR ANY DAMAGES, INCLUDING BUT NOT LIMITED TO SPECIAL, DIRECT, INDIRECT, INCIDENTAL OR CONSEQUENTIAL OR ANY LOSSES, SUCH AS BUT NOT LIMITED TO LOSS OF PROFIT, REVENUE, BUSINESS INTERRUPTION, BUSINESS OPPORTUNITY OR DATA THAT MAY ARISE FROM THE USE OF THIS DOCUMENT OR THE INFORMATION IN IT, EVEN IN THE CASE OF ERRORS IN OR OMISSIONS FROM THIS DOCUMENT OR ITS CONTENT.

Copyright and trademark: Nokia is a registered trademark of Nokia Corporation. Other product names mentioned in this document may be trademarks of their respective owners.

© 2026 Nokia.

Table of Contents

Summary of changes	7
1 Overview of the MantaRay HW Manager installation	10
2 Location and network connectivity	11
3 Prerequisites for installing the MantaRay HW Manager application	18
3.1 Hardware requirements	19
3.2 Software requirements	20
3.3 IP planning	21
3.4 MantaRay HW Manager host environment	24
3.4.1 Volumes and partitions	25
3.4.2 Storage capacity monitoring and alarming	27
3.4.3 MantaRay HW Manager security expectations from a host environment	27
3.5 Operating system	28
3.6 Installing supported OS	29
3.6.1 Configuring a default route	32
3.7 Creating a non-root user account	34
3.7.1 Creating a key for a non-root user	35
3.8 DNS configuration requirements	36
3.8.1 Checking the current host OS DNS configuration	38
3.8.2 Adding the nameserver IP address to the host OS DNS configuration	39
3.8.3 Adding the nameserver IP address to the host OS interface configuration	41
3.9 Configuring server time synchronization	44
3.9.1 Checking the chrony configuration	45
3.9.2 Configuring chrony on the MantaRay HW Manager host	46
3.9.3 Verifying the chrony configuration	47
3.10 Verifying MantaRay HW Manager package integrity	49
3.11 Prerequisites for installing the MantaRay HW Manager on OCP	51
4 Installing infrastructure	63
4.1 Installing the single-node infrastructure	63
4.2 Installing the three-node high availability infrastructure	71
5 Installing the MantaRay HW Manager application	80
5.1 Installing single-node local MantaRay HW Manager	81
5.2 Installing high availability local MantaRay HW Manager	89
5.3 Installing central MantaRay HW Manager	97
5.4 Installing the MantaRay HW Manager on OpenShift Container Platform	103
6 Post-installation actions	108
6.1 IP forwarding	112
6.2 Configuring log forwarding	114
6.2.1 Disabling log forwarding	116
6.3 Configuring login message in MantaRay HW Manager	117

7 Renewing Kubernetes certificates	121
7.1 Checking the Kubernetes certificates validity	121
7.2 Renewing Kubernetes certificates manually	124
7.2.1 Renewing valid Kubernetes certificates	124
7.2.2 Renewing expired Kubernetes certificates	129
8 Upgrading MantaRay HW Manager	136
8.1 Upgrading the MantaRay HW Manager on OCP	140
9 Removing MantaRay HW Manager and infrastructure	143
9.1 Removing MantaRay HW Manager	143
9.2 Removing infrastructure	144
9.3 Uninstalling the MantaRay HW Manager from OCP	145

List of Figures

Figure 1 MantaRay HW Manager network connectivity 11

Figure 2 Networking for high availability 16

Figure 3 MantaRay HW Manager installation options 18

Figure 4 IP planning in the high-availability configuration 22

Figure 5 MantaRay HW Manager with high availability and the dependent components
..... 29

Figure 6 Rocky Linux installation 31

Figure 7 Example output of the certificate validity check 111

Figure 8 View of the first login to MantaRay HW Manager 111

Figure 9 MantaRay HW Manager login message 118

List of Tables

Table 1	MantaRay HW Manager hardware requirements	20
Table 2	MantaRay HW Manager deployments size	25
Table 3	Capacity requirements of single-node local MantaRay HW Manager and high -availability local MantaRay HW Manager	27
Table 4	Parameters required for installing the single-node infrastructure	66
Table 5	Parameters required for installing the three-node high availability infrastructure	74
Table 6	MantaRay HW Manager application installation options	81
Table 7	Parameters required for installing local MantaRay HW Manager with the single -node infrastructure	83
Table 8	Parameters required for installing local MantaRay HW Manager with high availability	91
Table 9	Parameters required for installing central MantaRay HW Manager with the single -node infrastructure	99

Summary of changes

A list of changes between document issues. You can navigate through the respective changed topics.

[Changes between issues 19 \(2025-12-05, 26R1-NM\) and 20 \(2026-04-10, 26R2-NM\)](#)

General changes

The [Installing the MantaRay HW Manager on OpenShift Container Platform](#) has been added.

[Changes between issues 18 \(2025-08-14, 25R3-NM\) and 19 \(2025-12-05, 26R1-NM\)](#)

General changes

- The chapters [Creating a non-root user account](#) and [Creating a key for a non-root user](#) have been added.

Volumes and partitions

- [The paragraph](#) related to `/home/kubedata` has been updated.

Installing high availability local MantaRay HW Manager

- The `-K` parameter description has been updated in [Table: Parameters required for installing local MantaRay HW Manager with high availability](#).
- Username and password parameters examples have been aligned with other procedures.

Installing the single-node infrastructure

- *Before you start* section has been added.
- A [note](#) related to the `sudo` privileges has been updated.
- An [information](#) on the `-K` parameter has been updated in [Table: Parameters required for installing the single-node infrastructure](#). The table has been moved before the step examples.
- A [note](#) has been added to the step example. The `-U` and `-P` examples have been aligned with

other procedures.

Installing the three-node high availability infrastructure

- An information on the `-K` parameter has been updated in Table: Parameters required for installing the three-node high availability infrastructure. The table has been moved before the step examples.
- A note has been added to the step example. The `-U` and `-P` examples have been aligned with other procedures.

Installing single-node local MantaRay HW Manager

- Step examples have been updated. The password and parameters examples have been changed to be consistent with the Creating a non-root user account procedure.
- The example `-U` and `-P` parameters for the local account have been aligned with the ones created in the Creating a non-root user account procedure.
- The `-K` parameter description has been updated in Table: Parameters required for installing local MantaRay HW Manager with the single-node infrastructure.

Installing high availability local MantaRay HW Manager

- The `-K` parameter description has been updated in Table: Parameters required for installing local MantaRay HW Manager with high availability.
- Username and password parameters examples have been aligned with other procedures.

Installing central MantaRay HW Manager

- The `-K` parameter description has been updated in Table: Parameters required for installing central MantaRay HW Manager with the single-node infrastructure.
- Username and password parameters examples have been aligned with other procedures.

Post-installation actions

- The username and password examples have been aligned with other procedures.

IP forwarding

- The `sysctl.conf` changed to kernel tunables throughout the procedure.
- The location of this topic has been changed to the *Post-installation* section of the document.
- A new note on a default IP forwarding setting in Linux has been added.

Renewing valid Kubernetes certificates

- Procedure has been updated.

Renewing expired Kubernetes certificates

- Procedure has been updated.

Changes between issues 17 (2025-04-10, 25R2-NM) and 18 (2025-08-14, 25R3-NM)

Verifying MantaRay HW Manager package integrity

- This is a new chapter.

Installing the single-node infrastructure

- A step for verifying a digital signature file has been added.

Installing the three-node high availability infrastructure

- A step for verifying a digital signature file has been added.

Post-installation actions

- A clarification related to root and non-root user access has been added.

1. Overview of the MantaRay HW Manager installation

You can install MantaRay HW Manager on a single-node or a three-node infrastructure. The infrastructure you choose determines the type of MantaRay HW Manager application you can install.

Note:

Note that both *MantaRay HW Manager* and *Nokia AirFrame Data Center Manager (NADCM)* refer to the same product and are used interchangeably in the software and in the documentation.

MantaRay HW Manager installation contains two parts: the infrastructure and the MantaRay HW Manager application. The infrastructure consists of the Container Management framework with a container runtime management technology (such as Docker or containerd), Kubernetes, and support services.

This document describes the following issues:

- Infrastructures that you can choose for the MantaRay HW Manager application
- Types of the MantaRay HW Manager applications
- Correlation between the infrastructure and the type of the MantaRay HW Manager application
- Supported operating systems
- Post-installation actions
- Procedures needed to update and remove the MantaRay HW Manager application and infrastructure

2. Location and network connectivity

MantaRay HW Manager uses several types of connections and communication protocols.

Network connectivity

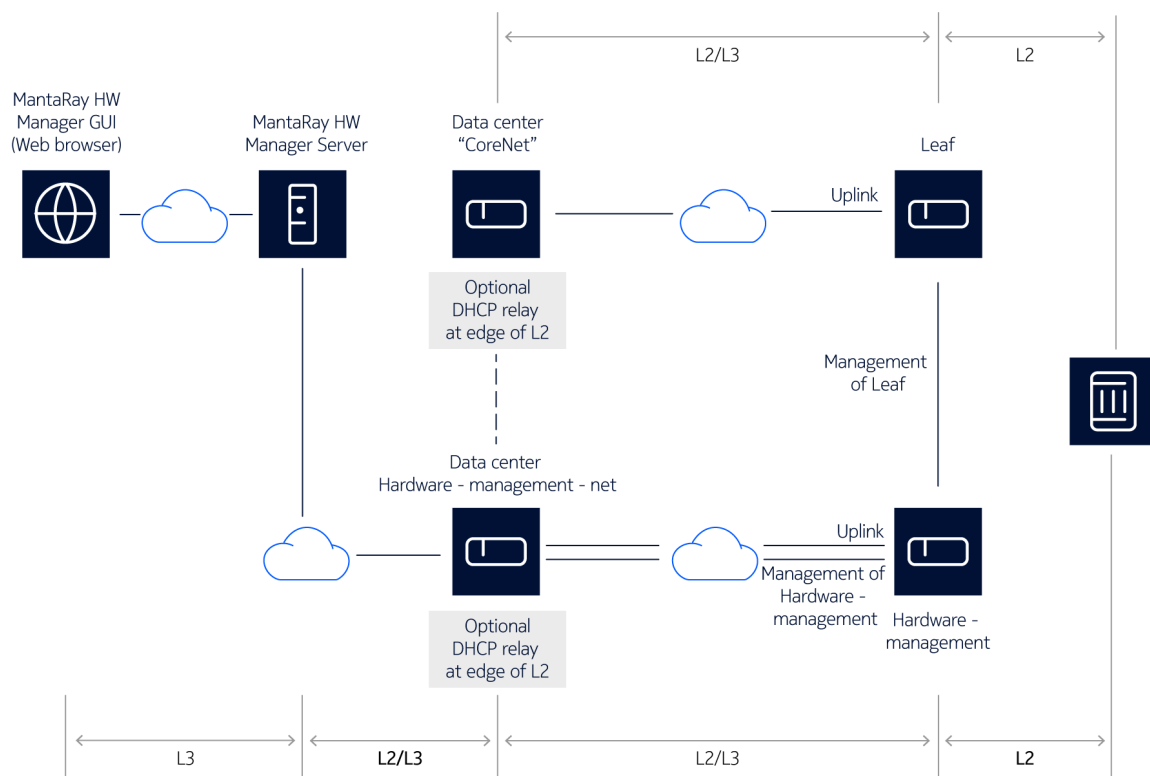
There are two connection types from and toward MantaRay HW Manager:

- User connection, for the management personnel to connect to MantaRay HW Manager
- Hardware management connection, for managing hardware

In addition to those two, if an external system is used to authenticate users, the connectivity to the authentication system exists.

When planning network configuration of the hardware, make sure that the firewall and edge router are configured correctly.

Figure 1: MantaRay HW Manager network connectivity



User connection

Connection is either from a web browser or through the REST application programming interface (API). Both connections are HTTPS traffic.

Management connection

Management connection can be either from the same or separate physical network interface than the user connection.

Management connectivity can be layer 2 (L2) or network layer (L3) connectivity.

DHCP

MantaRay HW Manager includes a Dynamic Host Configuration Protocol (DHCP) server, which is by default running but doesn't provide any service before configuring it. DHCP can provide addresses and, with the help of Trivial File Transfer Protocol (TFTP) or Hypertext Transfer Protocol (HTTP), boot images, and configuration files to the managed network nodes. The DHCP or TFTP services are provided to both L2 and L3 subnets. L3 subnets require a DHCP relay to be running at the edge of the L2 network.

If addresses are provided by MantaRay HW Manager or DHCP to the managed hardware, they can also be fixed at the managed nodes when requested. After the connectivity to the managed node is established, the user can request MantaRay HW Manager to alter the setting of a source IP address at the managed node (DHCP or fixed).

Protocols

Multiple protocols are used between MantaRay HW Manager and the managed nodes:

- Secure Shell (SSH)
- Remote Management Protocol (RMCP) + Intelligent Platform Management Interface (IPMI)
- Simple Network Management Protocol (SNMP)
- HTTPS (Redfish)

If DHCP is used, then also Bootstrap Protocol (BOOT) DHCP, TFTP, and HTTP are used.

Hardware management and leaf switches

The management of both switch types is done through their dedicated management interface. This is because certain software upgrades to switches require payload interfaces to be shut down.

AirFrame servers host network interfaces

Both Open Rack (OR) server and OpenEdge (OE) server have a shared LAN on motherboard (LoM) Network, Network Interface Controller (NIC) for management, and host which enables the host computer to be Preboot Execution Environment (PXE)-booted through the management network. Rackmount (RM) servers don't have a shared network interface card (NIC), therefore MantaRay HW Manager PXE booting host requires L3 (or L2) connectivity between MantaRay HW Manager and server host NIC.

Rack with factory settings

AirFrame Rack is delivered with an asset file. The asset file contains a set of nodes data, including their physical location in the rack. When an asset file with desired IP addresses is imported to MantaRay HW Manager, MantaRay HW Manager can assign those addresses to nodes and establish connectivity. The corresponding subnet has to be configured in MantaRay HW Manager and enabled.

Inspecting server nodes

Inspector can be run on server nodes to collect detailed hardware information. This is done by PXE booting the server node to a specific inspector agent.

Server provisioning

Desired operating system (OS) can be installed on server nodes with PXE booting.

Server deep inspection and provisioning rely on booting the server with the PXE procedure. The PXE procedure uses DHCP, TFTP, and HTTP protocols. Those protocols are sent and received by a server without Virtual Local Area Network (VLAN) tagging, so untagged traffic must be supported.

Note:

If a leaf switch is already configured to allow only tagged traffic on the ports toward servers, deep inspection and server provisioning can't work. To solve this, you need to reconfigure the leaf switch ports to a mode in which tagged and untagged traffic is allowed. After this, also untagged VLAN can be routed between MantaRay HW Manager and the servers.

The subnet address pool is dimensioned to allow servers to get addresses when doing deep inspection or provisioning. The process of getting the addresses consists of two stages:

1. Each server requests an address to get the boot image.
2. The OS requests an address to communicate with MantaRay HW Manager.

The received addresses are different as they come from other DHCP clients. A server consumes at least two IP addresses during deep inspection or provisioning. It means that:

- if requests from multiple server interfaces reach MantaRay HW Manager, even more addresses are assigned to a server.
- the same server gets different addresses when booting to the inspection agent multiple times as the inspection agent creates its *client identifier* during boot time using the current time.
- if the lease time for addresses is long and the address pool is small, all addresses may be leased, and the DHCP server can't assign an address to the client. This means the deep inspection or OS provisioning fails until the lease for the earlier reservations expires. For example, an address pool of 10 addresses with a (default) lease time of 3600s will allow a maximum of five servers to be deep inspected within one hour.

Note:

The lease time is configurable through the MantaRay HW Manager graphical user interface (GUI).

Networking for high availability mode

If you want to deploy MantaRay HW Manager with high availability, you need to deploy it on three nodes. The three nodes create a Kubernetes cluster. Deploying MantaRay HW Manager on a three-node cluster provides fault tolerance and high availability. When one of the nodes is unavailable, the other two take over its functions.

For MantaRay HW Manager, container technology (such as Docker or containerd) is used as a Platform-as-a-Service (PaaS) container management framework, used to develop and deliver software applications in packages called containers. The containers are isolated from other containers on the same OS. Each container runs its own software, configuration files, and libraries.

The containers are managed by Kubernetes. Kubernetes is an open-source container orchestration system for automating application deployment, scaling, and management. There are two main concepts in the Kubernetes cluster:

node A common term for bare-metal servers or virtual machines (VMs) that Kubernetes manages.

pod A basic unit of deployment in Kubernetes. A pod is a collection of containers that run on the same OS.

In MantaRay HW Manager, Kubernetes services act as an abstract layer for single or multiple pods. Kubernetes has load balancer IPs assigned for external incoming traffic. For internal communication, Kubernetes uses the internal cluster IP. The outgoing cluster traffic goes from an individual pod or a node (from the node where it serves the request) to the target.

The [Figure: Networking for high availability](#) presents an example of the deployment of MantaRay HW Manager on three nodes. The number of pods may differ in each node. You may find there are pods running the same instance on all three nodes.

There are two networks:

- Management network that connects the nodes with the managers, MantaRay NM, and MantaRay HW Manager GUI.
- Hardware management network that connects the nodes with the hardware.

There are two subnets:

- The northbound subnet with three IP addresses and one floating address that allow communication between the lower-level components and the higher-level components.

Note:

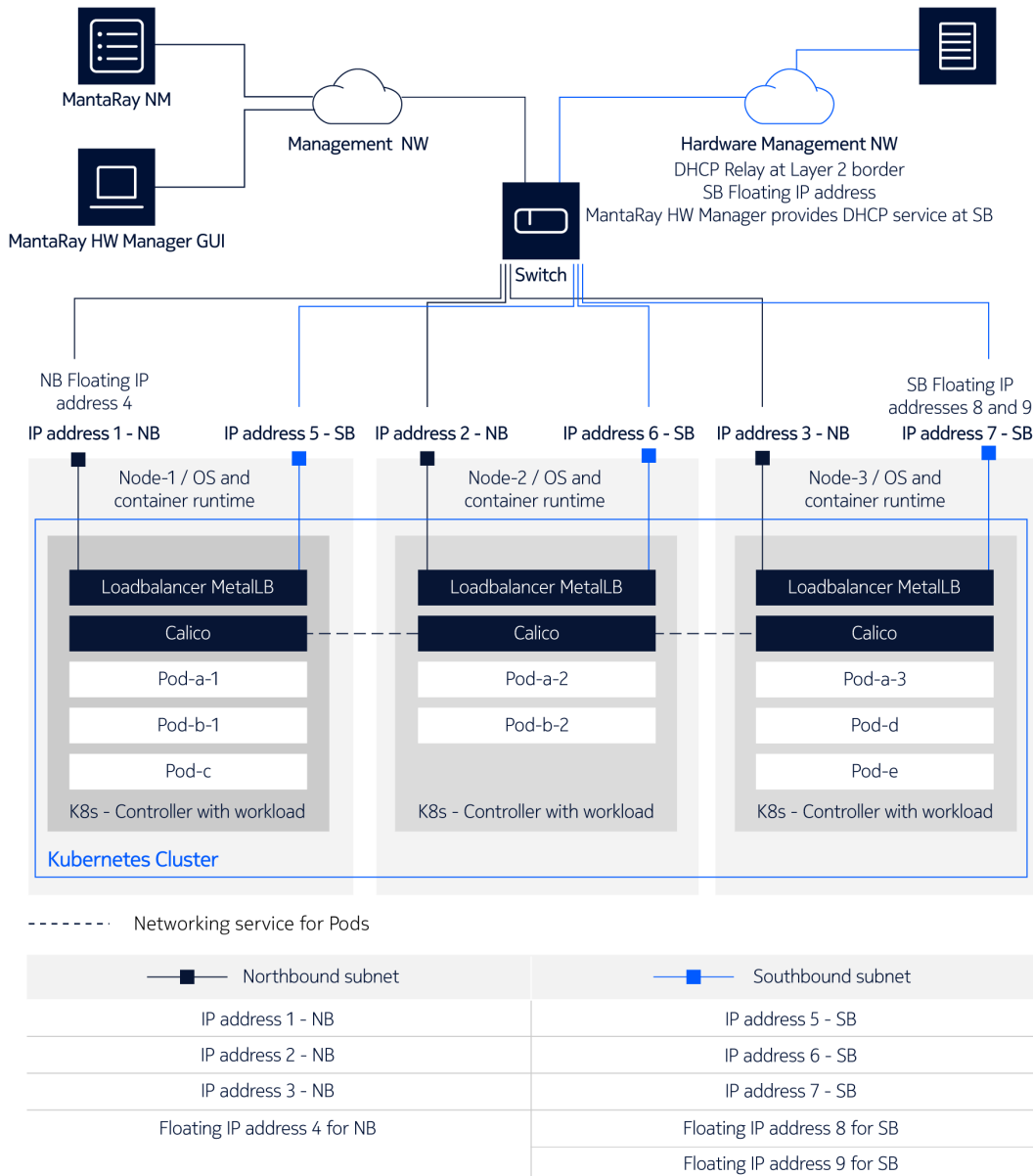
All four northbound addresses need to be in the same IP subnet.

- The southbound subnet with three IP addresses and two floating addresses that allow communication between the higher-level components and the lower-level components.



Note:
All five southbound addresses need to be in the same IP subnet.

Figure 2: Networking for high availability



HW

hardware

K8s

Kubernetes

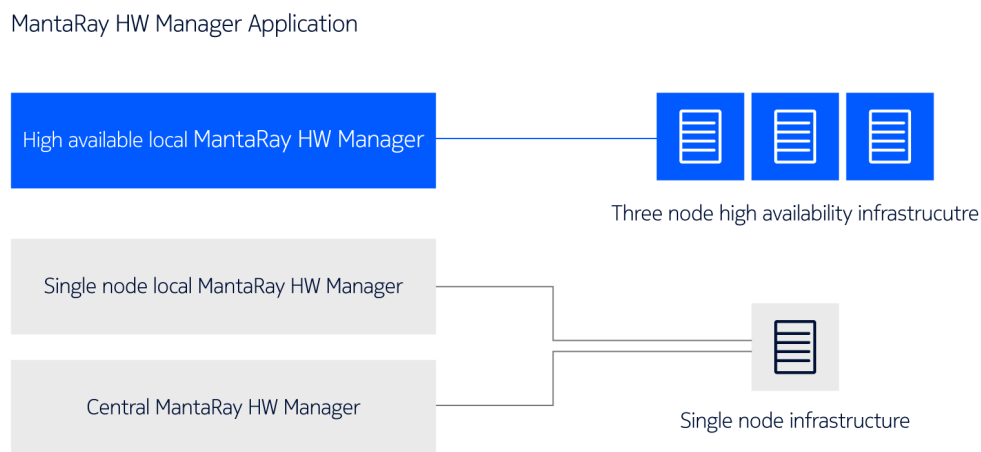
NW	network
NB	northbound
SB	southbound

3. Prerequisites for installing the MantaRay HW Manager application

The infrastructure you install during the MantaRay HW Manager setup determines the type of the MantaRay HW Manager application.

Before installing the MantaRay HW Manager application, make sure you have set up the appropriate infrastructure.

Figure 3: MantaRay HW Manager installation options



Notice:

You can't install the high-availability local MantaRay HW Manager on a single-node deployment.

If you have installed a single-node local MantaRay HW Manager application and you want to have a high-availability local MantaRay HW Manager application, you need to uninstall the infrastructure and install it in the three-node high-availability option.

Infrastructure

When installing the infrastructure, you can choose from two options:

- Single-node infrastructure that requires a single bare metal server or virtual machine (VM).
Nokia recommends the bare metal installation.

For more information, see [Installing the single-node infrastructure](#).

- Three-node high availability infrastructure that requires three bare metal servers.

For more information, see [Installing the three-node high availability infrastructure](#).

MantaRay HW Manager application

You can install the following options of the MantaRay HW Manager application:

- Single-node local MantaRay HW Manager that requires the single-node infrastructure
- Central MantaRay HW Manager that requires the single-node infrastructure
- High-availability local MantaRay HW Manager that requires the three-node high-availability infrastructure

For more information on the MantaRay HW Manager application options, see [Installing the MantaRay HW Manager application](#).

3.1 Hardware requirements

The hardware requirements are the same for the single-node MantaRay HW Manager and the three-node MantaRay HW Manager with high availability.

The following hardware requirements apply to both single-node MantaRay HW Manager and MantaRay HW Manager with high availability. They are valid for each individual host in the high-availability setup:

Table 1: MantaRay HW Manager hardware requirements

	Up to 50 nodes		51–300 nodes		301–1000 nodes		1001–7000 nodes	
	Minimum	Recommended	Minimum	Recommended	Minimum	Recommended	Minimum	Recommended
CPU (physical cores)	6	8	6	8	12	12	24	24
Memory (RAM GB)	32	64	64	128	64	128	128	256
Required storage capacity by MantaRay HW Manager	400 GB	500 GB (SSD)	500 GB	1 TB (SSD)	750 GB	1 TB (SSD)	1.5 TB (SSD)	2 TB (SSD)

The cores in the above recommendation refer to physical cores, not virtualized CPU (vCPU). For a virtualized deployment, 30% more vCPU capacity is recommended.

All hosts in the MantaRay HW Manager high availability setup need to be similar, including:

- CPU.
- memory.
- storage capacity.
- disk partitioning.
- host operating system (OS).
- network connections.

When dimensioning the host environment, consider the possible increase of node amount in the future. It is not necessarily straightforward to increase the host resources later on, in case the initially chosen specification turns out to be insufficient.

3.2 Software requirements

You need to make sure you have enough free user IDs and group IDs on the host to avoid overlapping permissions for the MantaRay HW Manager files.

MantaRay HW Manager reserves a range of Linux user IDs and group IDs on the host. The default

range for the user IDs and group IDs is 60000–60061.

If the IDs of either local users or users from Lightweight Directory Access Protocol (LDAP) conflict with the MantaRay HW Manager IDs usage, you need to re-configure and free the user IDs and group IDs.

3.3 IP planning

You need to create an IP plan for the environment before installing MantaRay HW Manager.

You can install MantaRay HW Manager in three modes:

IPv4-only In this mode, both the northbound and southbound interfaces support only IPv4.

IPv6-only In this mode, both the northbound and southbound interfaces support only IPv6.

Dual-stack In this mode, the northbound and southbound interfaces support IPv4 and IPv6.

Note:

You need to enable IPv6 in the host operating system (OS) regardless of IP installation mode, so the `ipv6.disable` parameter is 0.

Single-node configuration

In the single-node configuration, you need to plan the following IPs:

- Gateway IP
- Baseboard Management Controller (BMC) IP
- Northbound IP
- Southbound IP

Note:

All IPs need to be assigned and reachable.

Three-node high-availability configuration

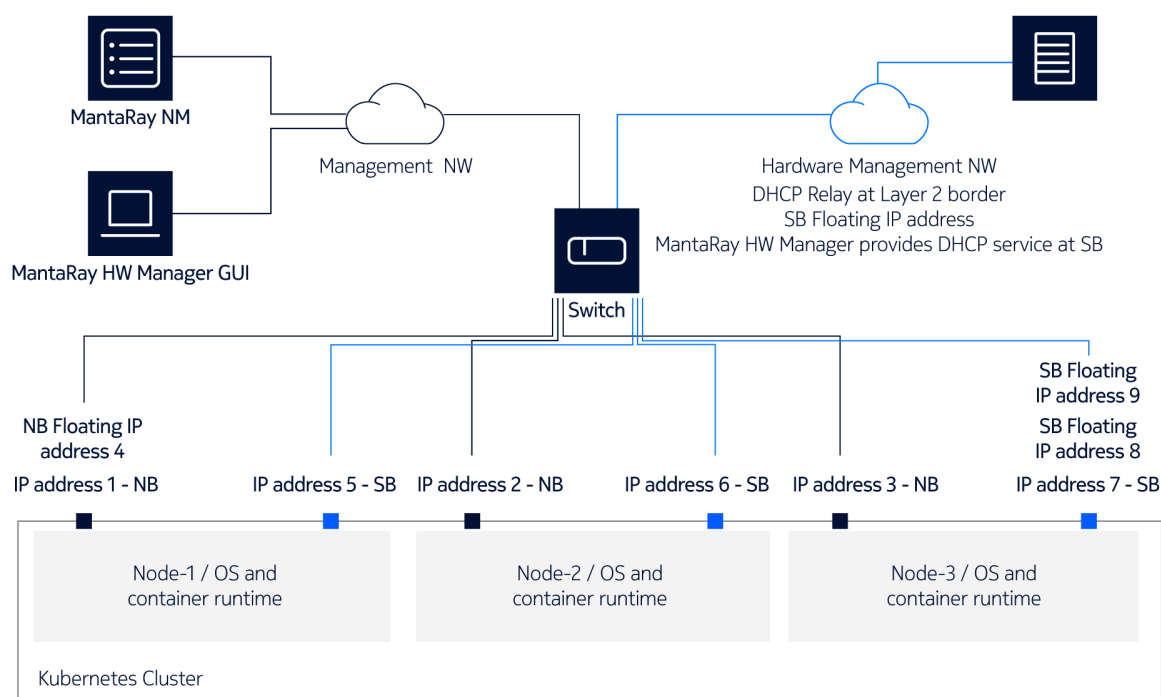
In the three-node high-availability IP planning, you need to plan the following IPs:

Tip:

You can check the IPs numbers in [Figure: IP planning in the high-availability configuration](#).

- Gateway IP
- BMC IP for all three servers used as MantaRay HW Manager hosts
- Host IP for all three servers used as MantaRay HW Manager hosts. See the northbound IP addresses 1-3 and southbound IP addresses 5-7.
- Northbound Floating IP. See the northbound floating IP address 4.
- Southbound Floating IP. See the southbound floating IP address 8.
- Simple Network Management Protocol (SNMP) trap receiver floating IP. See the southbound floating IP 9.

Figure 4: IP planning in the high-availability configuration



SB southbound

NB northbound

For details about MantaRay HW Manager with high availability, see [Figure: Networking for high availability](#).

While creating the IP plan for the three-node high-availability infrastructure installation, take into account the following details:

- You need to configure the host IP and the BMC IP before installing the three-node high availability configuration.
The IPs need to be assigned and reachable to all three servers.
- The northbound floating IP, southbound floating IP, and SNMP trap receiver floating IP need to be routable.
You can't attach those IPs to any interface.
- You can use the same IP address for northbound and southbound interfaces.

 Notice:

The SNMP trap receiver floating IP needs to differ from the southbound IP for the three-node high-availability configuration.

When setting the IP addresses for northbound and southbound interfaces (regardless of whether the addresses are the same or different), you need to check if a default route is set. If the route is missing, you can add it manually. For instructions, see [Configuring a default route](#).

Note:

For a single interface with a single northbound or southbound address, the default route needs to point to the gateway of the network to which the address belongs in the interface.

For two interfaces with separate addresses for the northbound or southbound interface:

- ▣ the default route needs to point to the gateway of the network to which the northbound address belongs in the northbound interface.
- ▣ a static route needs to be defined towards all target hardware (both BMC and machine networks) with the southbound interface.
- ▣ for additional security, Nokia recommends isolating northbound and southbound interfaces from each other by preventing packet forwarding between them.

For more information, see the *Security hardening* chapter in the *Configuring Security in MantaRay HW Manager* document.

Don't set a gateway IP to both interfaces if you use separate northbound and southbound addresses.

- If the initialize script uses the host URL as the northbound IP address, you need to use the MantaRay HW Manager hostname for the host URL.

3.4 MantaRay HW Manager host environment

You need to install the host environment with versions indicated in the MantaRay HW Manager Summary of Corrections and Enhancements document.

MantaRay HW Manager needs storage space for:

- Containers
- Firmware
- Operating system images
- Logs
- Databases
- Temporary files

The required storage space depends on the size of the managed devices and the number of images.

For the central and local MantaRay HW Manager hardware configuration, the size of the deployment depends on the number of the managed nodes:

Table 2: MantaRay HW Manager deployments size

Micro deployment	Small deployment	Medium deployment	Large deployment
<50 nodes	51-300 nodes	300-1000 nodes	1001-7000 nodes

3.4.1 Volumes and partitions

You need to ensure sufficient storage capacity before installing MantaRay HW Manager.

The operating system (OS) is not part of the MantaRay HW Manager delivery. The administrator of the MantaRay HW Manager host environment and its resources needs to ensure that the host performs adequately and that there are no capacity shortages in any partition of the storage.

Note:

For capacity requirements, refer to [Hardware requirements](#) and [Table: Capacity requirements of single-node local MantaRay HW Manager and high-availability local MantaRay HW Manager](#).

You need to partition the storage capacity according to the requirements of the MantaRay HW Manager application.

The information presented in [Table: Capacity requirements of single-node local MantaRay HW Manager and high-availability local MantaRay HW Manager](#) includes the OS capacity requirements but excludes the requirements of other applications (apart from Kubernetes and MantaRay HW Manager).

Note:

During the OS installation, the OS setup wizard adds the swap partitioning by default. However, Kubernetes recommends disabling the swap.

To disable the swap, remove it from the list of partitions during the OS installation process.

Kubernetes assumes a specific disk configuration. To simplify disk partitioning, MantaRay HW Manager assumes two important partitions: `root` and `home`. This applies to both single-node MantaRay HW Manager and MantaRay HW Manager with high availability. Ensure that the `root` and `home` partitions meet the capacity requirements specified in the [Table: Capacity requirements of single-node local MantaRay HW Manager and high-availability local MantaRay HW Manager](#) and that the following directories are located on the root partition (default behavior). The absence of some directories is expected on a newly installed operating system.

- `/var/log/pods`
- `/var/lib/containerd`
- `/var/lib/kubelet`
- `/etc/kubernetes/pki/`
- `/usr/bin/`
- `/usr/local/bin/`

Additionally, the application assumes that `/home/kubedata` can use 80% of the `/home` partition space. In case you require more space for other use in the `/home` partition, measure your disk and `/home` partition accordingly.

You need to partition the capacity according to the limits presented in the following table.

** Note:**

The table below describes the requirements for the MantaRay HW Manager application and OS. You need to have additional storage capacity to support other possible software.

Table 3: Capacity requirements of single-node local MantaRay HW Manager and high-availability local MantaRay HW Manager

Planned deployment size in terms of node amount	Up to 50 nodes (recommended only for single-node)	Up to 300 nodes	301-1000 nodes	1001-7000 nodes
Storage minimum size	400 GB	500 GB	750 GB	1.5 TB
Root partition (/) minimum size	250 GB	250 GB	250 GB	250 GB
Home partition (/home/) minimum size	150 GB	250 GB	450 GB	950 GB

3.4.2 Storage capacity monitoring and alarming

You can monitor the storage capacity in the MantaRay HW Manager graphical user interface (GUI).

In the MantaRay HW Manager GUI dashboard, you can view the usage percentage of the mountpoints and the raised active events in case of exceeded usage of the storage capacity.

Note:

Please pay attention to the *Disk Usage High* active events and remember to clean up unnecessary files. This will allow you to free up the capacity from the indicated mountpoint in case of raised events.

3.4.3 MantaRay HW Manager security expectations from a host environment

You need to harden the host security before installing MantaRay HW Manager.

To harden the host security, you need to implement the following practices:

- Follow the operating system (OS) hardening instructions where applicable:
 - Grant user access rights only to limited personnel that will perform the system

administration.

- Assign a unique static IP address to the host OS.
- Avoid disk partitioning, other than instructed in the Volumes and partitions chapter, to prevent inadvertent denials of service.
- Avoid installing other containers to the host where MantaRay HW Manager runs.
- Avoid any software that is not provided by Nokia. This is to make sure that no malicious software can interfere with the MantaRay HW Manager services.

Note:

If you apply the log policy in place, it guarantees that the system doesn't run out of vital resources and MantaRay HW Manager performs without interruptions.

- Place common security measures correctly. This includes, for example:
 - Intrusion Management
 - Intrusion Detection System (IDS)
 - Intrusion Prevention System (IPS)
 - Physical security appliances
- Harden layer 2 security configurations and ensure time synchronization.
For more information, see Configuring server time synchronization.
- Consider using a firewall in the southbound interface.

3.5 Operating system

MantaRay HW Manager supports the Red Hat Enterprise Linux (RHEL) or Rocky Linux operating systems (OS).

Note:

OS-related chapter is not applicable for installation on OpenShift Container Platform (OCP).

The MantaRay HW Manager installation packet doesn't include an OS, so you need to install it on the hardware you will use with MantaRay HW Manager.

MantaRay HW Manager supports the following OS:

- RHEL, which requires a commercial license for support and updates. You can contact Nokia for a commercial license. You can download RHEL from <https://www.redhat.com/en>.

- Rocky Linux, which is free open-source software. You can download it from <https://www.rockylinux.org/>.

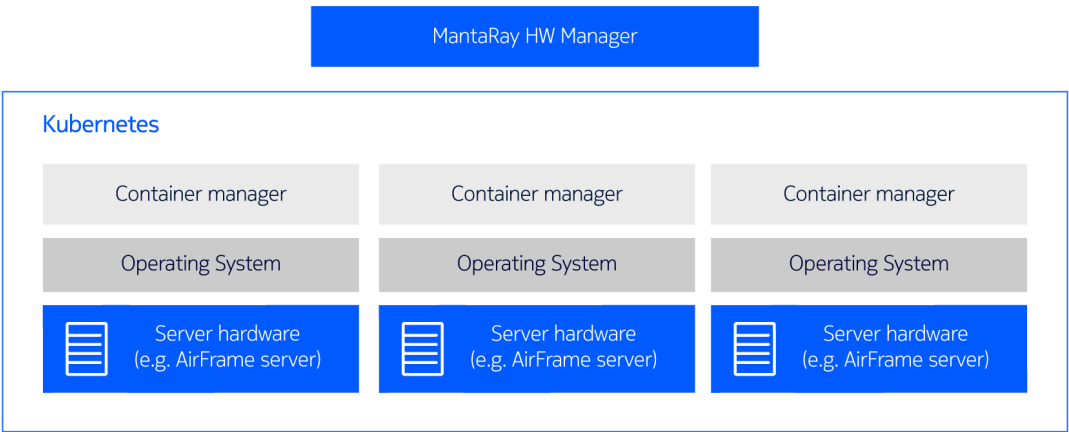
Note:

For information about the supported versions of the OS, see *MantaRay NM/Release Documentation/MantaRay HW Manager Summary of Corrections and Enhancements*.

After the installation, you need to install all security patches under the main release.

As the MantaRay HW Manager application runs in a set of containers, you can install MantaRay HW Manager on any device supporting the latest Kubernetes versions. Kubernetes schedules the containers to working servers. For example, it restarts failed containers and moves them to other physical servers in case of a server failure.

Figure 5: MantaRay HW Manager with high availability and the dependent components



3.6 Installing supported OS

You need to install Red Hat Enterprise Linux (RHEL) or Rocky Linux as an operating system (OS) on each of the MantaRay HW Manager hosts.

Before you start

Note:

OS-related chapter is not applicable for installation on OpenShift Container Platform (OCP).

Procedure

- 1 Download a supported OS.

For more information about the supported OS, see [Operating system](#).

Note:

The support for the base environment options between single-node MantaRay HW Manager and high-availability MantaRay HW Manager may vary. For the available options, see *MantaRay NM/Release Documentation/MantaRay HW Manager Summary of Corrections and Enhancements*.

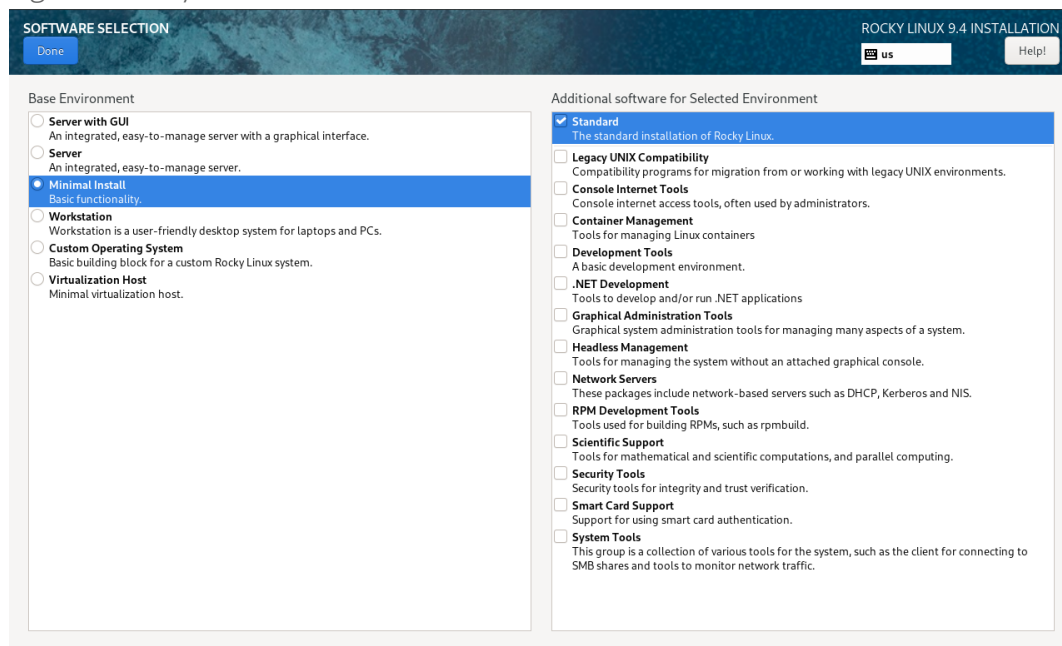
- 2 Begin the OS installation.
 - 2.1 In the `Software` selection window, select the `Minimal install` option.
 - 2.2 [For Rocky Linux] In the `Additional software for Selected Environment` section, select the `Standard` option.

Note:

The `Standard` option installs the `tar` package that you need to extract to the MantaRay HW Manager build.

Step example

Figure 6: Rocky Linux installation



- 3 In the installation destination, select **I will configure partitioning** and define a storage space for each partition.

For details on how to define storage space for partitions, see [Volumes and partitions](#).

Note:

When selecting a hard drive, Nokia recommends using an SSD.

- 4 Change the hostname.

Notice:

Don't change the hostname after you've installed MantaRay HW Manager. Altering the hostname after the MantaRay HW Manager installation results in incorrect functioning of Kubernetes and the MantaRay HW Manager application. If you need to change the MantaRay HW Manager host OS hostname after the installation process, you need to reinstall MantaRay HW Manager.

- 4.1 Run the following command:

```
sudo hostnamectl set-hostname <here-the-hostname-use-only-non  
-capital-letters>
```

4.2 Log out and log back in.

When you log back in, the hostname reflects correctly to the terminal session.

Step example

```
exit
```

Step result

You have changed the hostname.

Result

You have installed the OS.

3.6.1 Configuring a default route

To install MantaRay HW Manager, you need to set a default route.

Purpose

Setting a default route is needed as part of the operating system (OS) installation process. The route allows you to gain connectivity to a newly installed server.

Before you start

Note:

Make sure you set only one default route. The default route needs to always go through the northbound interface. Only static routes toward the managed device subnets are set for the southbound interface.

Procedure

- 1 Check a default route.

Step example

```
ip route | grep default  
default via 10.27.101.189 dev eth0
```

- 2 Check the connection.

Step example

```
sudo nmcli con
```

Step result

NAME	UUID	TYPE	DEVICE
System eth0	5fb06bd0-0bb0-7ffb-45f1-d6edd65f3e03	ethernet	eth0

The output needs to list all the available network connections and devices. System eth0 is used as an example of a connection name for the northbound network.

- 3 Add a default route.

Step example

For IPv4:

```
sudo nmcli con modify 'System eth0' ipv4.gateway 10.27.101.129
```

Step example

For IPv6:

```
sudo nmcli con modify 'System eth0' ipv6.gateway 2a00:8a00:4000:6009::b:f101
```

- 4 Check the default route again.

Step example

```
ip route | grep default  
default via 10.27.101.129 dev eth0
```

Result

You have configured a default route.

3.7 Creating a non-root user account

You need to create a non-root user with `sudo` access to proceed with the installation of MantaRay HW Manager.

Purpose

The user account and credentials created in this procedure are used in the examples that follow in the document. Use the account created in this procedure for downloading the software and performing the installation procedures.

Note:

You can change the credentials proposed in this procedure according to your needs, but remember about the change when using the examples provided in the document.

Before you start

Make sure the user account you are using can run the commands with `sudo` privileges.

Procedure

- 1 Create a local user account.

```
sudo useradd -m -G wheel nuser
```

- 2 Execute the following command to create a password for the `nuser` account.

```
sudo passwd nuser
```

- 3 Provide a new password.

Step example

In this document, the example `Pa55w#rd` password will be used.

Result

A new local account with `sudo` privileges is created.

Note:

Note down the username and password credentials for the created account. You will need them when performing other installation procedures.

Example

The following example credentials have been created and are used later in the document:

- user name: `nuser`
- password: `Pa55w#rd`

3.7.1 Creating a key for a non-root user

Instructions on how to create a Secure Shell (SSH) key for a non-root user

Purpose

You need to execute this procedure only if you want to use a key instead of a password for a local non-root user during the installation procedures.

Before you start

The `nuser` account is used in the steps below. If you created an account with a different

username during the [Creating a non-root user account](#) procedure, use that instead. In such a situation, remember to change `nuser` in the steps below to an actual account username.

Procedure

- 1 Create a passwordless key.

```
ssh-keygen -t ed25519 -f /home/nuser/.ssh/node.key
```

Press [Enter](#) twice to create a key with no passphrase.

- 2 Change file access rights to required.

```
chmod 600 /home/nuser/.ssh/node.key
```

- 3 Copy a public part of the key created to the host.

```
ssh-copy-id -i /home/nuser/.ssh/node.key nuser@10.37.44.185
```

Provide your non-root user password when asked.

Result

You have created the optional SSH key for the non-root user account. Use it instead of the password in further procedures.

3.8 DNS configuration requirements

You need to configure the host operating Domain Name System (DNS) to install the MantaRay HW Manager runtime components.

General requirements

Note:

In case of installation on OpenShift Container Platform (OCP), DNS configuration is performed as part of Network Management Common Cloud Platform (NMCCP) deployment.

Note:

For OCP deployments, FQDN resolution configuration must be performed as per OCP containers-as-a-Service (CaaS) specific instructions to ensure proper FQDN resolution.

Before installing MantaRay HW Manager, make sure that the `/etc/resolv.conf` file:

- is not completely empty.
- contains at least one DNS IP address.
- doesn't contain the `127.0.0.1` or the `localhost` DNS IP address.
- contains at least one IPv6 nameserver (in case of MantaRay HW Manager IPv6-only installation mode).

Note:

Keep in mind that if MantaRay HW Manager is installed with the fully qualified domain name (FQDN) as a host URL (`initialize script -l parameter`), you need to configure the nameserver, which is configured to the `/etc/resolv.conf` host, to resolve the used FQDN.

Running the DNS server on the same host as MantaRay HW Manager can lead to unpredictable configuration issues with Kubernetes services.

In case you don't fulfill these requirements, the Kubernetes internal name service fails to initialize correctly and prevents the MantaRay HW Manager installation.

To make sure all of the prerequisites are fulfilled, follow the procedures in the recommended order:

1. Check the current host operating system (OS) DNS configuration following the [Checking the current host OS DNS configuration](#) procedure.
2. Define the DNS configuration following the [Adding the nameserver IP address to the host OS DNS configuration](#) procedure.
3. Prevent network managers from making unwanted changes to the defined DNS configuration following the [Adding the nameserver IP address to the host OS interface configuration](#)

procedure.

Note:

During the MantaRay HW Manager installation, the installation script modifies the search and nameserver configuration in the host `/etc/resolv.conf` file. This is necessary for the Kubernetes cluster setup.

Re-installing infrastructure

When you re-install the infrastructure, remove any updates done by the previous infrastructure installation from the `/etc/resolv.conf` file.

Make sure you remove the following elements:

- the internal Kubernetes coreDNS nameserver
- the DNS search lines

```
search default.svc.cluster.local svc.cluster.local
nameserver 169.254.25.10
```

3.8.1 Checking the current host OS DNS configuration

You can check the current host operating system (OS) Domain Name System (DNS) from the `resolv.conf` file.

Procedure

- 1 Go to the `resolv.conf` file.

Step example

```
[root@nuser admin]# cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 10.171.0.1
[root@nuser admin]#
```

Step result

The DNS server addresses, if existing, are listed as `nameserver <ip_address>`.

Postrequisites

- If the `/etc/resolv.conf` file is empty, doesn't contain any nameserver addresses, or contains a nameserver with the `127.0.0.1` IP address, proceed with [Adding the nameserver IP address to the host OS DNS configuration](#).
- If the `/etc/resolv.conf` file contains at least one valid nameserver, you need to check if the DNS nameserver is defined also in the host OS interface configuration.
For more information, see [Adding the nameserver IP address to the host OS interface configuration](#).

Note:

Linux OS might have network services which automatically manage the DNS configuration based on the interface configuration of the system. This means that the entries currently present in the `resolv.conf` might, upon the next host reboot, be automatically overwritten if such a service is active and the interface configuration is not aligned with the `/etc/resolv.conf` file contents.

3.8.2 Adding the nameserver IP address to the host OS DNS configuration

You can add the Domain Name System (DNS) address to the `resolv.conf` file manually.

Before you start

- If the host operating system (OS) `/etc/resolv.conf` file doesn't contain any

nameserver addresses or is completely empty, add your DNS server address to the `resolv.conf` file.

- In case there are no valid DNS servers available in the network, you can define an artificial DNS address. For this purpose, use `192.0.2.2` which is from the IP address block that has been reserved for documentation purposes by Internet Engineering Task Force (IETF). It means that using this address should not create conflicts.

Note:

Don't use the `127.0.0.1` or `localhost` nameserver addresses. If these values are used, the Kubernetes name service fails to initialize. In case there are no valid DNS servers in the network, use an artificial IP address for the nameserver. Nokia recommends using the documentation block IP addresses, for example, `192.0.2.2` for IPv4 or `2001:db8:abcd::1` for IPv6.

Procedure

- 1 Add your DNS server address to the `resolv.conf` file.

- 1.1 Open the `resolv.conf` file.

Note:

You can edit files using the `vi` text editor.

Step example

```
vi /etc/resolv.conf
```

- 1.2 Add the nameserver IP addresses to the file.

```
nameserver 10.171.0.1
```

Note:

If you need to use an artificial DNS server address, don't use the `127.0.0.1` or `localhost` nameserver addresses.

**Note:**

If you install the IPv6-only mode, use the IPv6 DNS server address as the nameserver.

1.3 Save the changes.

2 Verify the modified DNS configuration.

Check if the changes made to the `resolv.conf` file are saved.

Step example

```
[root@nuser admin]# cat /etc/resolv.conf
nameserver 10.171.0.1
[root@nuser admin]#
```

For more information, see [Checking the current host OS DNS configuration](#).

Result

You have added the nameserver IP address to the host OS DNS configuration. You can see the nameserver in the `resolv.conf` file.

3.8.3 Adding the nameserver IP address to the host OS interface configuration

You need to check if the Domain Name System (DNS) address is configured properly in the host operating system (OS) interface configuration.

Purpose

Rocky Linux and Red Hat typically have NetworkManager service enabled by default. NetworkManager manages the host OS `/etc/resolv.conf` contents by populating them automatically with the information it reads from the interface configuration.

If none of the interfaces in the system has the DNS address configured, the `nameserver` IP address you have added manually to `/etc/resolv.conf` will be overwritten as an empty address during the next host reboot.

To make sure that the DNS addresses currently present in `/etc/resolv.conf` remain the same, check if the same DNS entries are also configured to the same host OS interface configuration. If they are not, add them to the host OS interface configuration.

Note:

You need to use the `nmcli` commands to add the nameserver to the host OS interface configuration.

Procedure

1 Check the interface configuration for the DNS entries.

- In IPv4-only and dual-stack installation mode, check the interface configuration over IPv4:

```
>> nmcli connection show eno1 | grep -i ipv4.dns
```

Step example

```
[root@nuser admin] >> nmcli connection show eno1 | grep -i ipv4.dns
ipv4.dns: 10.171.0.1,10.171.0.2,10.171.0.3
```

- In IPv6-only installation mode, check the interface configuration over IPv6:

```
>> nmcli connection show <interface-name> | grep ipv6.dns-search
```

Step example

```
[root@nuser admin] >> nmcli connection show eno1 | grep -i ipv6.dns
ipv6.dns: 2a00:8a00:5000:2f08::a,2a00:8a00:5000:2f08::b
```

Step result

- If the `nmcli` command shows the DNS addresses that are currently in the

`/etc/resolv.conf` file and are part of the interface configuration, you don't need to execute any other steps.

For example:

```
ipv4.dns: 10.171.0.1,10.171.0.2,10.171.0.3
```

- If the `nmcli` command doesn't show any values for `ipv4.dns` or `ipv6.dns`, and the DNS addresses haven't been defined in the host OS interface configuration, proceed to steps 2 and 3.

2 Add the DNS addresses to the interface configuration.

Note:

Configure the same `nameserver` IP addresses as in the `/etc/resolv.conf` file to one of the MantaRay HW Manager host server interfaces. Don't use `localhost` or `127.0.0.1`.

2.1 Select the interface of the configuration you want to edit.

Step example

- In IPv4-only and dual-stack installation mode:

```
[root@nuser admin]# nmcli c modify eno1 ipv4.dns
10.171.0.1,10.171.0.2,10.171.0.3 ipv4.dns-search
abc.xyz.my.net,xyz.my.net
```

- In IPv6-only installation mode:

```
[root@nuser admin]# nmcli c modify ens4f1 ipv6.dns
2a00:8a00:5000:2f08::a,2a00:8a00:5000:2f08::b ipv6.dns-search
abc.xyz.my.net,xyz.my.net
```

2.2 Restart the interface.

- In IPv4-only and dual-stack installation mode:

```
[root@nuser admin]# nmcli connection up eno1
```

- In IPv6-only installation mode:

```
[root@nuser admin]# nmcli connection up ens4f1
```

- 3 Check the `/etc/resolv.conf` file.

Step example

```
[root@nuser admin]# cat /etc/resolv.conf
```

Step result

The changes are visible in the `/etc/resolv.conf` file.

If you can't see the changes, reboot the host OS and re-check the `/etc/resolv.conf` file.

Result

You have added the nameserver IP address to the host OS interface configuration.

3.9 Configuring server time synchronization

This section describes how to configure the MantaRay HW Manager host to use `chrony` for time synchronization.

Note:

In case of installation on OpenShift Container Platform (OCP), server time synchronization configuration is performed as part of Network Management Common Cloud Platform (NMCCP) deployment.

Make sure that locally used time server IPs or fully qualified domain names (FQDNs) are available.

3.9.1 Checking the chrony configuration

You can check the status of the `chrony` configuration.

Procedure

- 1 Check the current status of the `chronyd` service.

```
# systemctl status chronyd
```

Step result

```
chronyd.service - NTP client/server
Loaded: loaded (/usr/lib/systemd/system/chronyd.service; disabled; vendor
preset: enabled)
Active: inactive (dead)
Docs: man:chronyd(8)
      man:chrony.conf(5)
```

- 2 Start `chronyd` (if it's not active).

```
# systemctl start chronyd
# systemctl enable chronyd
```

- 3 Track the current synchronization status.

```
chronyc tracking
```

Step result

```

Reference ID : 00000000 ( )
Stratum : 0
Ref time (UTC) : Thu Jan 01 00:00:00 1970
System time : 0.000000000 seconds slow of NTP time
Last offset : +0.000000000 seconds
RMS offset : 0.000000000 seconds
Frequency : 35.310 ppm slow
Residual freq : +0.000 ppm
Skew : 0.000 ppm
Root delay : 1.000000000 seconds
Root dispersion : 1.000000000 seconds
Update interval : 0.0 seconds
Leap status : Not synchronised

```

4 View the current synchronization sources.

```
chronyc sources
```

Step result

```

210 Number of sources = 0
MS Name/IP address Stratum Poll Reach LastRx Last sample
=====

```

Result

You have checked the `chrony` configuration, and the `chronyd` service is active and enabled.

3.9.2 Configuring chrony on the MantaRay HW Manager host

You can configure `chrony` on the MantaRay HW Manager host.

Procedure

- 1 Back up the `chrony` configuration file before making any modifications.

```
# cp /etc/chrony.conf /etc/chrony.conf-BU
```

- 2 Edit the `/etc/chrony.conf` file.

```
# vi /etc/chrony.conf
```

- 3 Add your local time reference server information to the `/etc/chrony.conf` file.

- 3.1 Comment out the current servers from the beginning of the file using `#`.

```
#pool 2.rhel.pool.ntp.org iburst
```

- 3.2 Add your local server information using a fully qualified domain name (FQDN) or IP address.

```
server <IP-address1> iburst
server <IP-address2> iburst
```

- 4 Restart the `chronyd` service.

```
# systemctl restart chronyd
```

Result

You have configured `chrony` on the MantaRay HW Manager host. To verify the configuration, go to [Verifying the chrony configuration](#).

3.9.3 Verifying the chrony configuration

You can check if `chrony` is successfully configured.

Procedure

- 1 Check the current status of the `chronyd` service.

```
# systemctl status chronyd
```

Step result

The output is as follows:

```
systemd[1]: Starting NTP client/server...
chronyd[1695]: chronyd version 3.5 starting (+CMDMON +NTP +REFCLOCK +RTC
+PRIVDROP +SCFILTER +SIGND +ASYNCDNS +SECHASH +IPV6 +DEBUG)
chronyd[1695]: Frequency -35.310 +/- 49.810 ppm read from
/var/lib/chrony/drift
chronyd[1695]: Using right/UTC timezone to obtain leap second data
systemd[1]: Started NTP client/server.
chronyd[1695]: Selected source <IP of the selected server>
chronyd[1695]: System clock TAI offset set to 37 seconds4
```

2 Check the current synchronization status.

```
chronyc tracking
```

Step result

The output is as follows:

```
Reference ID : C264318B (FQDNS or IP of connected reference server)
Stratum : 3
Ref time (UTC) : Fri Aug 27 10:50:10 2021
System time : 0.000004440 seconds slow of NTP time
Last offset : +0.000014202 seconds
RMS offset : 0.000485875 seconds
Frequency : 34.803 ppm slow
Residual freq : -0.013 ppm
Skew : 1.209 ppm
Root delay : 0.017485864 seconds
Root dispersion : 0.001125142 seconds
Update interval : 65.0 seconds
Leap status : Normal
```

3 View the current synchronization sources.

```
chronyc sources
```

Step result

The output is as follows:

```
210 Number of sources = 2
MS Name/IP address Stratum Poll Reach LastRx Last sample
=====
^* (IP/FQDNS) 2 6 377 61 -104us[ -100us] +/- 10ms
^+ (IP/FQDNS) 2 6 377 61 -190us[ -190us] +/- 8878us
```

Result

You have successfully configured chrony.

3.10 Verifying MantaRay HW Manager package integrity

In Software Supply Tool (SWST), you can find the MantaRay HW Manager release package, a checksum file, and a digital signature file. The latter is signed with an official Nokia root certificate authority (CA) certificate that you can use to verify the integrity of the MantaRay HW Manager package.

Before you start

1. Go to SWST on [Nokia Support Portal](#) and download the digital signature file.
2. Download the Nokia root CA certificate from the following link: <https://nokia.certificate.fi/>.

Example:

```
curl -k0 https://nokia.certificate.fi/crt/Nokia_Root_CA_Certificate.crt
```

Note:

Make sure that the host machine has access to the URL for downloading Nokia certificates.

Procedure

- 1 Convert the Nokia root CA certificate to PEM format using `openssl`.

```
openssl x509 -inform DER -in Nokia_Root_CA_Certificate.crt -out rootca.crt
```

- 2 Extract the signature ZIP file.

```
unzip nadcm_full-<release>.tgz.sha256_SIGNATURE.zip
```

- 3 Verify the signature with the PEM-formatted Nokia root CA certificate using `openssl`.

Note:

Note that the actual hash might differ from the example below.

```
openssl smime -verify -in hash_container_hwmanager.sig -inform DER -content  
hash_container_hwmanager.txt -CAfile rootca.crt -purpose any
```

Step result

Example of a successful output:

```
e5df971d8daac70ac6aa734c753a3fca3ba4aa12339aa4c9cda09aee1008aa95  
nadcm_full-<release>.tgz  
Verification successful
```

Example of an unsuccessful output (damaged `hash_container_hwmanager.txt` file):

```
aaaaa71d8daac70ac6aa734c753a3fca3ba4aa12339aa4c9cda09aee1008aa95  
nadcm_full-<release>.tgz  
Verification failure  
800B9285FB7F0000:error:10800065:PKCS7 routines:PKCS7_signatureVerify:digest  
failure:crypto/pkcs7/pk7_doit.c:1096:  
800B9285FB7F0000:error:10800069:PKCS7 routines:PKCS7_verify:signature  
failure:crypto/pkcs7/pk7_smime.c:363:
```

- 4 To guarantee integrity, calculate the release package hash locally and compare it with the signature verification output from step 3.

```
sha256sum nadcm_full-<release>.tgz
e5df971d8daac70ac6aa734c753a3fca3ba4aa12339aa4c9cda09aee1008aa95
nadcm_full-<release>.tgz
```

Result

You have verified the MantaRay HW Manager package integrity.

3.11 Prerequisites for installing the MantaRay HW Manager on OCP

Before installing the MantaRay HW Manager application on OpenShift Container Platform (OCP), make sure you have set up the appropriate environment.

Cluster availability

Ensure the OCP cluster is available and accessible before beginning the MantaRay HW Manager installation.

Hardware and software requirements

See the [Prerequisites for installing the MantaRay HW Manager application](#) for hardware, software requirements and planning, as the same prerequisites apply for MantaRay HW Manager installation on OCP clusters.

OpenShift CLI (oc) tool

The oc tool must be present on the installer node to:- Generate authentication tokens for OCP cluster login- Authenticate to the OCP internal registry

You can verify the oc tool availability by using the `which oc` and `oc version` commands.

```
# which oc
/usr/local/bin/oc
```

```
# oc version
Client Version: 4.14.3
Kustomize Version: v5.0.1
Server Version: 4.14.3
Kubernetes Version: v1.27.3
```

Kubectl CLI tool

`kubectl` command-line tool must be present on the installer/provisioner node. This is required for interacting with the cluster, executing administrative tasks, and deploying applications.

Use the `which kubectl` to verify whether `kubectl` is installed and accessible on your installer/provisioner node, and the `kubectl version` to check the installed version.

```
# which kubectl
/usr/bin/kubectl
```

```
# kubectl version
Client Version: v1.28.3
Kustomize Version: v5.0.4
Server Version: v1.28.2
```

Kubectl configuration

`kubectl` must be configured with the proper kubeconfig file to access the target OCP cluster.

1. Set the kubeconfig file used for the current session:

```
# export KUBECONFIG=/home/nmsocpadmin/clusterconfigs/auth/kubeconfig
```

2. Verify cluster access:

```
# kubectl get pods -A
```

NAMESPACE	NAME	READY	STATUS	
	RESTARTS	AGE		
kube-system	coredns-565d847f94-abcde	1/1	Running	0
	12d			
default	my-app-7c9f8d6c4f-xyz12	1/1	Running	1
	3h			

Note:

The result is a list of all pods in the OpenShift-related namespaces.

Container manager

podman is highly recommended for pushing images to the OCP internal or external registry.

1. Set podman as the container manager:

```
# export ContainerManager=podman
```

2. Verify podman installation:

```
# podman --version
podman version 4.4.1
```

Storage class

A storage class must be configured in the OCP cluster for persistent volume claims.

1. Retrieve available storage classes:

```
# kubectl get storageclasses
```

NAME	PROVISIONER	RECLAIMPOLICY	
VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE	
standard (default)	kubernetes.io/aws-ebs	Delete	
WaitForFirstConsumer	true	45d	
fast	kubernetes.io/gce-pd	Retain	Immediate
false	20d		
slow	kubernetes.io/no-provisioner	Retain	Immediate
false	15d		

2. Set the OCPSTORAGECLASS environment variable:

```
# export OCPSTORAGECLASS="<your-chosen-storage-class-name>"
```

Replace the **<your-chosen-storage-class-name>** with the actual name of the StorageClass you intend to use, for example:

```
# export OCPSTORAGECLASS="standard"
```

Internal OCP registry configuration

Before pushing images to OCP internal registry, make sure:

- You have access to an OCP cluster.
- OpenShift CLI (oc) tool is installed.
- You are logged in as a user with cluster-admin or registry-editor permissions.
- You have an authentication token for registry login.

1. Generate a token to log in to the registry, by either:

- Using an oc command:

```
# oc login -u <username> -p <password>
https://api.<cluster_name>.<base_domain>:6443
Login successful.
```

You have access to 5 projects, the list has been saved in
"/home/user/.kube/config".

Using project "default".

You can use the `oc whoami -t` command to output the authentication token for the currently logged-in user.

- Or, by using the web-based token retrieval, by navigating and logging in to:

```
https://oauth-openshift.apps.<cluster_name>.<base_domain>/oauth/token/request
```

For example:

```
https://oauth-openshift.apps.slab723cluster.slopc.net.com/oauth/token/request
```

And setting the:

- OCP internal or external registry route:

```
# export IMAGEREGISTRY_ADDRESS=default-route-openshift-image-registry.apps.<cluster_name>.<base_domain>
```

example:

```
# export IMAGEREGISTRY_ADDRESS=default-route-openshift-image-registry.apps.cluster01.example.com
```

- Image repository name in the registry:

```
# export IMAGEREGISTRY_REPOSITORY=hwm
```

- Cluster admin username with push/pull permissions:

```
# export IMAGEREGISTRY_USERNAME=<cluster-admin username>
```

- Cluster admin password:

```
# export IMAGEREGISTRY_PASSWORD=<cluster-admin password>
```

2. Setting the following parameters:

- OCP internal or external registry route:

```
# export IMAGEREGISTRY_ADDRESS=default-route-openshift-image-registry.apps.<cluster_name>.<base_domain>
```

example:

```
# export IMAGEREGISTRY_ADDRESS=default-route-openshift-image
-image-registry.apps.clab3835.ds.netact.nsn-rdnet.net
```

Tip:

The internal OCP registry address can be retrieved from the cluster oc routes by picking the value of HOST field (oc login must be done prior to executing any oc commands):

```
# oc get routes -A
```

NAMESPACE	NAME	HOST/PORT
PATH SERVICES	PORT TERMINATION	WILDCARD
openshift-image-registry	default-route	default-route-openshift-image-registry.apps.clab3835.ds.netact.nsn-rdnet.net
image-registry	<all> reencrypt	None

- Image repository name in the registry:

```
# export IMAGEREGISTRY_REPOSITORY=hwm
```

- Cluster admin username with push/pull permissions:

```
# export IMAGEREGISTRY_USERNAME=<cluster-admin username>
```

- Token from oc login command:

```
# export IMAGEREGISTRY_PASSWORD=<token from oc login command>
```

External OCP registry configuration

You can use an external registry (for example: Harbor or central container registry (CCR)) by setting the following environment variables:

- Harbor ingress address:

```
# export IMAGEREGISTRY_ADDRESS=<harbor ingress>
```

For example:

```
# export IMAGEREGISTRY_ADDRESS=nmccp.sprintlab905.netact.nsn-rdnet.net
```

💡 Tip:

The CCR harbor ingress address can be retrieved from the OCP cluster configuration by picking the HOST:

```
# oc get ingress -A
```

NAMESPACE	NAME	CLASS	HOSTS
ADDRESS		PORTS	AGE
cal-shared-product	ccr-harbor-ingress	nmapps-ingress	
nmccp.sprintlab905.netact.nsn-rdnet.net			
100.84.2.18,2a00:8a00:4000:20c::24d:12	80, 443	37d	

- Image repository name in the registry:

```
# export IMAGEREGISTRY_REPOSITORY=hwm
```

- Robot account name (push/pull access to hwm project):

```
export IMAGEREGISTRY_USERNAME=nmccphwm
```

💡 Tip:

This user will be newly created within the NMCCP harbor registry. Therefore, to access images in the project, the username must contain the *nmccp* prefix.

- Robot account password example:

```
# export IMAGEREGISTRY_PASSWORD="Nokia123!"
```

- CCR Harbor admin user password example (for creating project):

```
# export CCR_PASSWORD="Nokia123!"
```

Image handling options

Specify a custom image temporary directory:

```
export IMGTMPDIR=/data/hwm/images
```

Note:

If `IMGTMPDIR` is not set, the default temporary directory is `/tmp`.

The minimum required capacity of the `/tmp` directory (or the directory set in `IMGTMPDIR`) is 10 GB.

MetalLB configuration (Multi-Node OCP)

For MNO (Multi-Node OpenShift) clusters, MetalLB must be installed and configured before HWM installation.

IP requirements

MantaRay HW Manager requires at least one virtual IP on the northbound interface and two virtual IPs on the southbound interface.

MetalLB operator installation

Make sure MetalLB operator is installed and MetalLB instance is created in `metallb-system` namespace.

Use the following command to verify if MetalLB is installed and working:

```
kubectl get all -n metallb-system
```

Example of command output where MetalLB is installed and running:

NAME	READY	STATUS	RESTARTS	AGE
pod/controller-7c7b8f6f6d-9xqkz	1/1	Running	0	12d
pod/speaker-2h9gk	1/1	Running	0	12d
pod/speaker-7f8md	1/1	Running	0	12d
pod/speaker-kxq5w	1/1	Running	0	12d

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/metallb-webhook	ClusterIP	10.96.123.45	<none>	443/TCP	12d

NAME	DESIRED	CURRENT	READY	UP-TO-DATE	AVAILABLE
AGE					
deployment.apps/controller	1	1	1	1	1
12d					

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/controller-7c7b8f6f6d	1	1	1	12d

NAME	DESIRED	CURRENT	READY	UP-TO-DATE	AVAILABLE	AGE
daemonset.apps/speaker	3	3	3	3	3	12d

See the OCP documentation for details on MetalLB installation and configuration: [↗](#)

https://docs.redhat.com/en/documentation/openshift_container_platform/4.18/html/networking_operators/metallb-operator.

IP address pools and L2Advertisement configuration

Create IP address pools for northbound and southbound networks and L2Advertisement resources for each IP address pool.

1. Create a configuration file, for example: `metallb_config_ochwm1.yaml` with the following content:

```

apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
  name: northbound
  namespace: metallb-system
spec:
  addresses:
    - 10.91.83.171-10.91.83.171
    -
2a00:8a00:4000:020c:0000:0000:0007:0406-2a00:8a00:4000:020c:0000:0000:0007:04
06
  autoAssign: true
---
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
  name: southbound
  namespace: metallb-system
spec:
  addresses:
    - 10.91.84.47-10.91.84.47
    -
2a00:8a00:4000:020c:0000:0000:0007:0489-2a00:8a00:4000:020c:0000:0000:0007:04
89
  autoAssign: false
---
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
  name: snmptarget
  namespace: metallb-system
spec:
  addresses:
    - 10.91.87.123-10.91.87.123
    -
2a00:8a00:4000:020c:0000:0000:0007:0b15-2a00:8a00:4000:020c:0000:0000:0007:0b
15
  autoAssign: falseapiVersion: metallb.io/v1beta1

```

```

---
kind: L2Advertisement
metadata:
  name: northbound
  namespace: metallb-system
spec:
  ipAddressPools:
  - northbound
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: southbound
  namespace: metallb-system
spec:
  ipAddressPools:
  - southbound
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: snmptarget
  namespace: metallb-system
spec:
  ipAddressPools:
  - snmptarget

```

2. Apply the configuration to the OCP:

```
# oc apply -f metallb_config_ochwm1.yaml
```

3. You can use the following commands to verify:

- the IP address pools:

```
# kubectl get ipaddresspools.metallb.io -A
```

- the L2 advertisements:

```
# kubectl get l2advertisements.metallb.io -A
```

OVN-Kubernetes configuration

The OVN network configuration must be modified to allow pods to communicate directly with external devices on secondary interfaces. If the pods cannot ping external devices on the secondary network interface, then traffic egresses only through the primary interface, regardless of routing table configuration. In such a case, routing needs to be enabled through the host by modifying the cluster network operator configuration using the following command:

```
# oc edit network.operator/cluster
```

and adding or modifying the following settings for `defaultNetwork`:

```
spec:
  defaultNetwork:
    type: OVNKubernetes
    ovnKubernetesConfig:
      egressIPConfig: {}
      gatewayConfig:
        ipForwarding: Global
        ipv4: {}
        ipv6: {}
      routingViaHost: true
```

Pods may restart after the changes are applied. You can verify the configuration using the following command:

```
# oc get network.operator cluster -o yaml | grep routingViaHost
```

Additional network requirements

- Firewall rules must allow communication between the installer node and OCP nodes.
- Domain Name System (DNS) resolution must be configured for the OCP cluster's fully qualified domain name (FQDN) and routes.

4. Installing infrastructure

You need to install either the single-node or three-node high availability configuration before installing the MantaRay HW Manager application.

Note:

Installing infrastructure is not applicable for installation on OpenShift Container Platform (OCP).

When installing the infrastructure, you need to choose an appropriate configuration for the chosen MantaRay HW Manager application. You can install:

- single-node deployment for the single-node local MantaRay HW Manager and the central MantaRay HW Manager.
- three-node high availability deployment for the high availability local MantaRay HW Manager.

For more information, see [Table: MantaRay HW Manager application installation options](#).

Notice:

You can't install the high availability local MantaRay HW Manager on the single-node deployment.

If you have installed the single-node local MantaRay HW Manager application and you want to have the high availability local MantaRay HW Manager application, you need to uninstall the infrastructure and install it as the three-node high availability option.

4.1 Installing the single-node infrastructure

You can install the single-node local MantaRay HW Manager and the central MantaRay HW Manager application on the single-node infrastructure.

Before you start

The examples in this procedure use the local account credentials created in the [Creating a non](#)

-root user account procedure:

- username: nuser
- password: Pa55w#rd

Replace those credentials with the actual username and password for your local account when performing the procedure.

Procedure

- 1 Log in as a non-root user.

Note:

Use the `sudo -l` command to ensure that the non-root user's local account can perform the steps below with `sudo` privileges.

- 2 Use the `# hostnamectl` command to check if the hostname is other than `localhost`.

If the `static hostname` value of the command output contains `localhost`, `localhost.localdomain`, or similar, follow the instructions in the [Installing supported OS](#) chapter.

Note:

The Kubernetes certificate update doesn't work if the hostname is set to `localhost`. In such a case, the installation stops working after one year.

- 3 Check that the needed mounts exist and there is enough free space in the partitions.

```
df -h / /home
```

For more details, see [Volumes and partitions](#).

- 4 Download MantaRay HW Manager from the [Support Portal](#).

- 5 Verify a digital signature file with the Nokia root certificate authority (CA) certificate.

For instructions, see [Verifying MantaRay HW Manager package integrity](#).

Note:

If you have trouble getting the Nokia root CA certificate, go to step 6. Make sure that the checksum files have remained unaltered.

6 [Optional] Verify the integrity of the downloaded package.

Note:

Perform this step only if you haven't performed step 5.

```
sha256sum -c <checksum file> (e.g. sha256sum -c nadcm_full
-<release>.tgz.sha256)
```

7 Extract the release.

```
tar zxvf nadcm_full-<release>.tgz
```

8 Install and configure the infrastructure.

The MantaRay HW Manager application requires the following components:

- Containerd—container runtime
- Kubernetes—container orchestrator
- Helm 3—package manager for orchestrator
- Cert-manager

You can install the required infrastructure components using the `./initialize-infra` script.

Note:

In the examples below, you need to use either the user Secure Shell (SSH) login password (`-P` parameter) or the user SSH key (`-K` parameter).

Step example

Table 4: Parameters required for installing the single-node infrastructure

Parameter name	Description
-H, --hosts	This parameter is mandatory and defines the list of hosts. - For a baremetal high availability configuration, the list of IP addresses needs to be separated with a space. These IP addresses are from the northbound subnet. - For a single-node configuration, only one IP address is needed. - For a single-node virtual machine (VM) configuration where the external IP address isn't visible inside the VM, this value is the internally visible IP address. Example: <pre>-H "10.20.30.50 10.20.30.51 10.20.30.52"</pre>
-U, --user	This parameter is mandatory and defines the user who logs in to the nodes and runs sudo commands.
-P, --password	This parameter defines the password for the user who logs in to the nodes and runs sudo commands. If it's not provided, the user needs to have the passwordless sudo on the node. Note: Remember to introduce the password in single quotes.
-K, --ssh-key	This parameter defines the path to the key to use for SSH key-based authentication. If not provided, password (-P) is used for authentication. You can test the ssh connectivity with the key using the following command : <pre>ssh <user>@<host> -i <key_file></pre> Example: <pre>ssh nuser@10.20.30.50 -i /home/nuser/.ssh/node.key</pre> Note: The path above is an example. The key can be stored anywhere on the host. Make sure you have the right file permission, at least 600: <pre>chmod 600 /home/<user>/.ssh/node.key</pre>
-M, --install-mode	This parameter is mandatory and defines the installation mode: IPv4, IPv6, or dual-stack.

Parameter name	Description
-N, --northbound	This parameter is mandatory and defines the northbound IP address. • For a baremetal high availability configuration, the load balancer northbound floating IP address is used. • For a baremetal single-node configuration, the northbound IP address can be the same as the -H parameter. • For a single-node VM configuration, this value is the externally visible IP address. Example: -N 10.20.30.53
-S, --southbound	This parameter defines the southbound IP address. If set to default, it's the same as the northbound IP address. • For a baremetal high availability configuration, the load balancer southbound floating IP address is used. • For a baremetal single-node configuration, it's the IP address in the interface that the managed devices use for communication. • For a single-node VM configuration, this value is the externally visible IP address that the managed devices use for communication. Example: -S 192.168.1.40
-T, --snmp	This parameter is mandatory for a high availability infrastructure installation. It's the load balancer IP address to receive Simple Network Management Protocol (SNMP) traps. • This IP address needs to be different than the southbound address. • This parameter isn't used for a single-node, baremetal, or VM configuration. Example: -T 192.168.1.41
-N6, --northbound6	This parameter is optional and defines the northbound IPv6 address. • For a baremetal high availability configuration, the load balancer northbound floating IPv6 address is used. • For a baremetal single-node configuration, the northbound IP address can be the same as the host IPv6 address. • For a single-node VM configuration, this value is the externally visible IP address. Example: -N6 fd00:dcde:0:61::a053

Parameter name	Description
-S6, --southbound6	This parameter is optional and defines the southbound IPv6 address. If set to default, it's the same as the northbound IPv6 address. • For a baremetal high availability configuration, the load balancer southbound floating IPv6 address is used. • For a baremetal single-node configuration, it's the IPv6 address in the interface that the managed devices use for communication. • For a single-node VM configuration, this value is the externally visible IPv6 address that the managed devices use for communication. This parameter needs to be in a compressed format, for example: <pre>-S6 fd00:deed:acdc:1908::a053</pre>
-T6, --snmp6	This parameter is optional for a high availability infrastructure installation. It's the load balancer IPv6 address to receive SNMP traps. • This IP address needs to be different than the southbound IPv6 address. • This parameter isn't used for a single-node, baremetal, or VM configuration. This parameter needs to be in a compressed format, for example: <pre>-T6 fd00:deed:acdc:1908::a054</pre>
-h, --help	This parameter displays the help message and the available options.

Note:

The path `chmod 600 /home/<user>/.ssh/node.key` is only an example. The key can be stored anywhere on the host. Also, the IP addresses used in the examples below are only the examples. The `-U` and `-P` parameters should correspond to the user name and password created in the [Creating a non-root user account](#) procedure.

■ IPv4-only infrastructure installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra -H "10.37.44.138" \
-U nuser -P 'Pa55w#rd' \
-N 10.37.44.138 -S 192.168.1.43 \
-M ipv4
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra -H "10.37.44.138" \
-N 10.37.44.138 -S 192.168.1.43 \
-U nuser -K "/home/nuser/.ssh/node.key" \
-M ipv4
```

■ IPv6-only infrastructure installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra -H 2001:dcde:0:61::50 \
-U nuser -P 'Pa55w#rd' \
-N6 2001:dcde:0:61::50 \
-S6 fd00:deed:acdc:1908::50 \
-M ipv6
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra -H 2001:dcde:0:61::50 \
-N6 2001:dcde:0:61::50 \
-S6 fd00:deed:acdc:1908::50 \
-U nuser -K "/home/nuser/.ssh/node.key" \
-M ipv6
```

Note:

Remember that the `-N6` and `-S6` addresses should be entered in a compressed format.

■ Dual-stack infrastructure installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra -H "10.37.44.138" \
-U nuser -P 'Pa55w#rd' \
-N 10.37.44.138 -S 192.168.8.50 \
-N6 2001:dcdc:0:61::50 \
-S6 fd00:deed:acdc:1908::50 \
-M dual
```

With the user SSH key (the `-K` parameter):

```
ccd nadcm-<release>/
sudo ./initialize-infra -H "10.37.44.138" \
-N 10.37.44.138 -S 192.168.8.50 \
-N6 2001:dcdc:0:61::50 \
-S6 fd00:deed:acdc:1908::50 \
-U nuser -K "/home/nuser/.ssh/node.key" \
-M dual
```

Note:

Remember that the `-N6` and `-S6` addresses should be entered in a compressed format.

Result

You have installed the single-node infrastructure.

PLAY RECAP

```
*****
*****
*****
node1          : ok=3    changed=1    unreachable=0    failed=0
skipped=0      rescued=0    ignored=0
Install log created to '/home/nuser/nadcm-22FP2-IB8-22.9.1/cubeha
-infra-0.2.6/20220914_1559_install-infra.log'
```

4.2 Installing the three-node high availability infrastructure

You can install the local MantaRay HW Manager with high availability on the three-node high availability infrastructure.

Before you start

If you want to install the infrastructure in IPv6-only installation mode, you need to have the IPv4 address disabled on the interface.

Nokia recommends configuring time synchronization to make sure that certificates used internally by Kubernetes are valid for each host in the high availability configuration. For instructions, see [Configuring server time synchronization](#). Without time synchronization, the MantaRay HW Manager installation may fail or the Kubernetes cluster may function incorrectly in the long run.

Procedure

- 1 Log in as a non-root user.

Note:

Ensure the non-root user can run commands with `sudo`. You need the same user account on all three hosts where MantaRay HW Manager is installed.

- 2 Use the `# hostnamectl` command to check if the hostname is different than `localhost`.

If the command output is `localhost`, `localhost.localdomain`, or similar, follow [step 4. Change the hostname](#) instructions from the [Installing supported OS](#) chapter.

Note:

The Kubernetes certificate update doesn't work if the hostname is set to `localhost`. In such case, the installation stops working after one year. Make sure to define the hostname for all three nodes.

- 3 Check that the needed mounts exist and there is enough free space in the partitions for each

of the three nodes.

```
df -h / /home
```

For more details, see [Volumes and partitions](#).

- 4 Download MantaRay HW Manager from the [Support Portal](#).
- 5 Verify a digital signature file with the Nokia root certificate authority (CA) certificate.

For instructions, see [Verifying MantaRay HW Manager package integrity](#).

Note:

If you have trouble getting the Nokia root CA certificate, go to step 6. Make sure that the checksum files have remained unaltered.

- 6 [Optional] Verify the integrity of the downloaded package.

Note:

Perform this step only if you haven't performed step 5.

```
sha256sum -c <checksum file> (e.g. sha256sum -c nadcm_full
-<release>.tgz.sha256)
```

- 7 Extract the release.

```
tar zxvf nadcm_full-<release>.tgz
```

- 8 Install and configure the infrastructure.

The MantaRay HW Manager application requires the following components:

- Containerd—container runtime
- Kubernetes—container orchestrator
- Helm 3—package manager for orchestrator
- Cert-manager

You can install the required infrastructure components using the `./initialize-infra`

script.

Note:

In the examples below, you need to use either the user Secure Shell (SSH) login password (`-P` parameter) or the user SSH key (`-K` parameter).

Step example

Table 5: Parameters required for installing the three-node high availability infrastructure

Parameter name	Description
-H, --hosts	This parameter is mandatory and defines the list of hosts. - For a baremetal high availability configuration, the list of IP addresses needs to be separated with a space. These IP addresses are from the northbound subnet. - For a single-node configuration, only one IP address is needed. - For a single-node virtual machine (VM) configuration where the external IP address isn't visible inside the VM, this value is the internally visible IP address. Example: <pre>-H "10.20.30.50 10.20.30.51 10.20.30.52"</pre>
-U, --user	This parameter is mandatory and defines the user who logs in to the nodes and runs sudo commands.
-P, --password	This parameter defines the password for the user who logs in to the nodes and runs sudo commands. If it's not provided, the user needs to have the passwordless sudo on the node.  Note: Remember to introduce the password in single quotes.
-K, --ssh-key	This parameter defines the path to the key to use for SSH key-based authentication. If not provided, password (-P) is used for authentication. You can test the ssh connectivity with the key using the following command : <pre>ssh <user>@<host> -i <key_file></pre> Example: <pre>ssh nuser@10.20.30.50 -i /home/nuser/.ssh/node.key</pre>  Note: The path above is an example. The key can be stored anywhere on the host. Make sure you have the right file permission, at least 600: <pre>chmod 600 /home/<user>/.ssh/node.key</pre>
-M, --install-mode	This parameter is mandatory and defines the installation mode: IPv4, IPv6, or dual-stack.

Parameter name	Description
-N, --northbound	This parameter is mandatory and defines the northbound IP address. • For a baremetal high availability configuration, the load balancer northbound floating IP address is used. • For a baremetal single-node configuration, the northbound IP address can be the same as the -H parameter. • For a single-node VM configuration, this value is the externally visible IP address. Example: -N 10.20.30.53
-S, --southbound	This parameter defines the southbound IP address. If set to default, it's the same as the northbound IP address. • For a baremetal high availability configuration, the load balancer southbound floating IP address is used. • For a baremetal single-node configuration, it's the IP address in the interface that the managed devices use for communication. • For a single-node VM configuration, this value is the externally visible IP address that the managed devices use for communication. Example: -S 192.168.1.40
-T, --snmp	This parameter is mandatory for a high availability infrastructure installation. It's the load balancer IP address to receive Simple Network Management Protocol (SNMP) traps. • This IP address needs to be different than the southbound address. • This parameter isn't used for a single-node, baremetal, or VM configuration. Example: -T 192.168.1.41
-N6, --northbound6	This parameter is optional and defines the northbound IPv6 address. • For a baremetal high availability configuration, the load balancer northbound floating IPv6 address is used. • For a baremetal single-node configuration, the northbound IP address can be the same as the host IPv6 address. • For a single-node VM configuration, this value is the externally visible IP address. This parameter needs to be in a compressed format, for example: -N6 fd00:dcde:0:61::a053

Parameter name	Description
-S6, --southbound6	This parameter is optional and defines the southbound IPv6 address. If set to default, it's the same as the northbound IPv6 address. • For a baremetal high availability configuration, the load balancer southbound floating IPv6 address is used. • For a baremetal single-node configuration, it's the IPv6 address in the interface that the managed devices use for communication. • For a single-node VM configuration, this value is the externally visible IPv6 address that the managed devices use for communication. This parameter needs to be in a compressed format, for example: <pre>-S6 fd00:deed:acdc:1908::a053</pre>
-T6, --snmp6	This parameter is optional for a high availability infrastructure installation. It's the load balancer IPv6 address to receive SNMP traps. • This IP address needs to be different than the southbound IPv6 address. • This parameter isn't used for a single-node, baremetal, or VM configuration. This parameter needs to be in a compressed format, for example: <pre>-T6 fd00:deed:acdc:1908::a054</pre>

Note:

The path `chmod 600 /home/<user>/.ssh/node.key` is only an example. The key can be stored anywhere on the host. Also, the IP addresses used in the examples below are only the examples. The `-U` and `-P` parameters should correspond to the user name and password created in the [Creating a non-root user account](#) procedure.

■ IPv4-only infrastructure installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra -H "10.37.44.138 10.37.44.139 10.37.44.140" \
-U nuser -P 'Pa55w#rd' -M ipv4 -N 10.37.44.185 -S 192.168.1.43 \
-T 192.168.1.41
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra -H "10.37.44.138 10.37.44.139 10.37.44.140" \
-U nuser -K "/home/nuser/.ssh/node.key" -M ipv4 -N 10.37.44.185 -S
192.168.1.43 \
-T 192.168.1.41
```

■ IPv6-only infrastructure installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra \
-H "2a00:8a00:4000:312a::50 2a00:8a00:4000:312a::51
2a00:8a00:4000:312a::52" \
-U nuser -P 'Pa55w#rd' -M ipv6 -N6 2a00:8a00:4000:312a::a053 \
-S6 fd00:deed:acdc:1908::a053 -T6 fd00:deed:acdc:1908::a054 \
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra \
-H "2a00:8a00:4000:312a::50 2a00:8a00:4000:312a::51
2a00:8a00:4000:312a::52" \
-U nuser -K "/home/nuser/.ssh/node.key" -M ipv6 -N6
2a00:8a00:4000:312a::a053 \
-S6 fd00:deed:acdc:1908::a053 -T6 fd00:deed:acdc:1908::a054 \ -K
"/home/nuser/.ssh/node.key" \
```

Note:

Remember that the `-N6` and `-S6` addresses should be entered in a compressed format.

■ Dual-stack infrastructure installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra -H "10.37.44.138 10.37.44.139 10.37.44.140" \
-U nuser -P 'Pa55w#rd' -M dual -N 10.37.44.185 -S 192.168.8.53 \
-T 192.168.8.54 -N6 2001:dcdc:0:61::a053 -S6 fd00:deed:acdc:1908::a053 \
-T6 fd00:deed:acdc:1908::a054
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize-infra -H "10.37.44.138 10.37.44.139 10.37.44.140" \
-U nuser -K "/home/nuser/.ssh/node.key" -M dual -N 10.37.44.185 -S
192.168.8.53 \
-T 192.168.8.54 -N6 2001:dcdc:0:61::a053 -S6 fd00:deed:acdc:1908::a053 \
-T6 fd00:deed:acdc:1908::a054
```

Note:

Remember that the `-N6` and `-S6` addresses should be entered in a compressed format.

Result

You have installed the three-node high availability infrastructure.

Example: Three-node high availability infrastructure installation

PLAY RECAP

```
*****
*****
*****
```

```
node1          : ok=2    changed=1    unreachable=0    failed=0
skipped=0      rescued=0    ignored=0
node2          : ok=2    changed=1    unreachable=0    failed=0
skipped=0      rescued=0    ignored=0
node3          : ok=2    changed=1    unreachable=0    failed=0
skipped=0      rescued=0    ignored=0
```

```
Install log created to '/home/nuser/nadcm-NADCM22FP1/cubeha
-infra-0.2.2/20220530_0814_install-infra.log'
```

5. Installing the MantaRay HW Manager application

You can install the MantaRay HW Manager application in three different variants.

 Notice:

For installing MantaRay HW Manager in the Nokia Edge Automation Tool (NEAT) application, see *Nokia Edge Automation Tool/Operating Documentation/Installing NEAT*.

You can choose from three variants of the MantaRay HW Manager application presented in the following table:

Table 6: MantaRay HW Manager application installation options

MantaRay HW Manager application	Description
Single-node local MantaRay HW Manager	• Requires the single-node infrastructure • Automates the hardware management • Out-of-band interface to the managed hardware • Supports IPv4, IPv6, and dual-stack installation modes
High availability local MantaRay HW Manager	• Requires the high availability three-node infrastructure • Automates hardware management • Out-of-band interface to the managed hardware • Supports IPv4, IPv6, and dual-stack installation modes
Central MantaRay HW Manager	• Requires the single-node infrastructure • Gathers information about all local MantaRay HW Manager applications connected to the central MantaRay HW Manager • Displays the monitoring of all local MantaRay HW Manager applications connected to the central MantaRay HW Manager • Offers a limited set of control functions through the local MantaRay HW Manager applications • Supports IPv4, IPv6, and dual-stack installation modes

Note:

You can't install the single-node local MantaRay HW Manager and central MantaRay HW Manager on the same host.

5.1 Installing single-node local MantaRay HW Manager

You can install the single-node local MantaRay HW Manager application only on the single-node infrastructure.

Procedure

- 1 Log in as a non-root user.

Note:

Ensure that the non-root user can run commands with `sudo`. The same user account is needed on all three hosts, where MantaRay HW Manager is installed.

- 2 Start the local MantaRay HW Manager installation.

```
sudo ./initialize
```

Note:

In case your installation size exceeds 3000 nodes, remember to specify the `xxl` size option. For more information, see the `-z`, `--size` parameter in the following table.

Tip:

To see possible command-line arguments for the installation, use the following command:

```
./initialize -h
```

Table 7: Parameters required for installing local MantaRay HW Manager with the single-node infrastructure

Parameter name	Description
-l, --hosturl	This parameter is mandatory, required for the northbound URL. The host URL allows access to the MantaRay HW Manager graphical user interface (GUI) from a browser. Example of the --hosturl value: - For IPv4 installation: -l https://10.37.44.185 - For IPv6 installation, you need to provide the IPv6 address in a compressed format: -l https://[2001:dcdc:0:61::50] - For dual-stack installation: -l https://nadm50.dyn.nesc.nokia.net The fully qualified domain name (FQDN) is mandatory for dual-stack installation, however, you can also use FQDN in IPv4 and IPv6 installation modes.
-s, --south-ip	This parameter is mandatory, required for the southbound address. This IP address is used to manage hardware devices. You need to provide this parameter for IPv4 or dual-stack installation modes.
-t, --snmp-ip	This parameter is mandatory and defines the address of the MantaRay HW Manager Simple Network Management Protocol (SNMP) trap listener. You need to provide this parameter for IPv4 or dual-stack installation modes.
-u, --auth-admin	This parameter defines the name for the admin user account. The account is created in the authentication system (Keycloak) during the installation. If you don't specify this parameter, the admin default value is used.
-p, --auth-pass	This parameter defines the password for the admin user. If you don't specify it, a random password is generated during the installation and displayed to the user. Note: Remember to introduce the password in single quotes.
-e, --admin-email	This parameter defines the e-mail address of the admin user. If you don't specify the address, the <i>automatic@nokia.com</i> default value is used.

Parameter name	Description
-w, --hw-manager-user	This parameter defines the initial user of MantaRay HW Manager. The MantaRay HW Manager user account for the MantaRay HW Manager northbound login is created during the installation. If you don't specify the parameter, the <i>autogenerated</i> default value is used.
-n, --hw-manager-pass	This parameter defines the password for the initial MantaRay HW Manager user. If you don't specify the parameter, a random password is generated during the installation and displayed. Note: Remember to introduce the password in single quotes. You are required to change this password the first time you log in to MantaRay HW Manager.
-U, --username	This parameter is mandatory for the user on the Kubernetes node. The parameter concerns the local user on each of the nodes and requires sudo permission.
-P, --password	This parameter defines the local user password on each of the nodes. Note: Remember to introduce the password in single quotes.
-K, --ssh-key	This parameter defines the path to the key to use for SSH key-based authentication. If not provided, password (-P) is used for authentication. You can test ssh connectivity with the key using the following command : <pre>ssh <user>@<host> -i <key_file></pre> Example : <pre>ssh nuser@10.20.30.50 -i /home/user/.ssh/node.key</pre> Note: The path above is an example. The key can be stored anywhere on the host. Make sure you have the right file permission, at least 600: <pre>chmod 600 /home/<user>/.ssh/node.key</pre>

Parameter name	Description
<code>-s6, --south-ip6</code>	This parameter defines the southbound IPv6 address. It's required for IPv6 or dual-stack installation. You need to provide the IPv6 address in a compressed format.
<code>-t6, --snmp-ip6</code>	This parameter defines the SNMP IPv6 address. It's required for IPv6 or dual-stack installation. You need to provide the IPv6 address in a compressed format.
<code>-q, --sb-fqdn</code>	This parameter defines the southbound FQDN. It adds FQDN to the southbound subject alternative name (SAN) to allow the registry usage with the name. It's required for installing Nokia Edge Automation Tool (NEAT) in IPv6 mode. Don't use this parameter with standalone MantaRay HW Manager. Example: <code>-q my.southbound.neat.net</code>
<code>-z, --size</code>	This parameter is optional and defines the planned MantaRay HW Manager installation size. The size refers to the estimated maximum amount of managed devices. If you provide this parameter, the <code>xxL</code> size value is applicable. This is the value of an xx-large MantaRay HW Manager installation, which is up to 7000 managed devices. Specifying this parameter is needed to ensure that the <code>xxL</code> size works correctly. The specified size indicates a sufficient amount of host resources (for example, memory) that needs to be requested by the MantaRay HW Manager Kubernetes services. If the requested memory requirements aren't fulfilled in the host, for example, if the value is <code>-z xxL</code> , the installation is stopped.
<code>-h, --help</code>	This parameter displays the help message and the available options.

Run the `./initialize` script with the following parameters. The parameters presented in the example:

- create an admin user to the authentication system.
- create a password for the admin user.
- define an e-mail address for the admin user.

- create a user in MantaRay HW Manager.
- create a password for the MantaRay HW Manager user.
- define the southbound IP address and the host URL where MantaRay HW Manager is running.

Note:

In the examples below, you need to use either the user SSH login password (`-P` parameter) or the user SSH key (`-K` parameter).

Step example

- IPv4-only installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://10.37.44.185 -s 192.168.1.43 -t 192.168.1.43 \
-u admin -p 'P@ssw0rd' -e automatic@nokia.com -w nadcmadmin -n 'P@55w0rd' \
-U nuser -P 'Pa55w#rd'
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://10.37.44.185 -s 192.168.1.43 -t 192.168.1.43 \
-u admin -p 'P@ssw0rd' -e automatic@nokia.com -w nadcmadmin -n 'P@55w0rd' \
-U nuser -K "home/nuser/.ssh/node.key"
```

- IPv6-only installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://[2001:dcdc:0:61::50] \
-s6 fd00:deed:acdc:1908::50 \
-t6 fd00:deed:acdc:1908::50 \
-u admin -p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' -U nuser -P 'Pa55w#rd'
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://[2001:dcdc:0:61::50] \
-s6 fd00:deed:acdc:1908::50 \
-t6 fd00:deed:acdc:1908::50 \
-u admin -p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' -U nuser -K "home/nuser/.ssh/node.key"
```

Note:

Remember that the `-s6` and `-t6` addresses should be entered in a compressed format.

■ Dual-stack installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://nadcm50.dyn.nesc.nokia.net -s 192.168.8.50 \
-t 192.168.8.50 -s6 fd00:deed:acdc:1908::50 -t6 fd00:deed:acdc:1908::50 \
-u admin -p 'P@ssw0rd' -e automatic@nokia.com -w nadcmadmin -n 'P@55w0rd' \
-U nuser -P 'Pa55w#rd'
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://nadcm50.dyn.nesc.nokia.net -s 192.168.8.50 \
-t 192.168.8.50 -s6 fd00:deed:acdc:1908::50 -t6 fd00:deed:acdc:1908::50 \
-u admin -p 'P@ssw0rd' -e automatic@nokia.com -w nadcmadmin -n 'P@55w0rd' \
-U nuser -K "home/nuser/.ssh/node.key"
```

Note:

In the case of the dual-stack installation, remember that:

- ▣ the `-l` parameter needs to be an FQDN. For example:
https://nadcm.dyn.mydomain.net.
- ▣ the northbound and the southbound interfaces need to be dual-stack.
- ▣ the `-s6` address should be entered in a compressed format.

Replace the IP addresses of the `-s`, `-l`, and `-t` parameters with the ones applicable to your environment.

Step result

Expected results (the IP and password will be different according to the site):

```
***** BINARY INTEGRITY *****
```

Note! Execute binary integrity check again using script:

```
./scripts/binary_integrity_check.sh
```

Note! Release package binary integrity can be checked manually (without script 'binary_integrity_check.sh') by executing in command line:

```
sha256sum -c --quiet binary_integrity/release_files/nadcm*
```

(result shall notify from changed or missing files. New files are not noted.)

To setup authentication or to add new users:

```
https://10.21.20.252/auth/admin/
```

```
username: admin
```

```
password: P@ssw0rd
```

To login to MantaRay HW Manager as an autogenerated user:

```
https://10.21.20.252
```

```
username: nadcmadmin
```

```
password: P@55w0rd
```

NOTE: Please store all the credentials above carefully

Result

You have installed the single-node local MantaRay HW Manager application. You can proceed to Post-installation actions.

5.2 Installing high availability local MantaRay HW Manager

You can install the high availability local MantaRay HW Manager application only on the three-node high availability infrastructure.

Procedure

- 1 Log in as a non-root user.

Note:

Ensure that the non-root user can run commands with `sudo`. The same user account is needed on all three hosts, where MantaRay HW Manager is installed.

2 Start the local MantaRay HW Manager installation.

```
sudo ./initialize
```

Note:

In case your installation size exceeds 3000 nodes, remember to specify the `xxl` size option. For more information, see the `-z`, `--size` parameter in the following table.

Tip:

To see possible command-line arguments for the installation, use the following command:

```
./initialize -h
```

Table 8: Parameters required for installing local MantaRay HW Manager with high availability

Parameter name	Description
-l, --hosturl	This parameter is mandatory, required for the northbound URL. The host URL allows access to the MantaRay HW Manager GUI from a browser. Example of the --hosturl value: - For IPv4 installation: -l https://10.37.44.185 - For IPv6 installation, you need to provide the IPv6 address in a compressed format: -l https://[2001:dcde:0:61::50] - For dual-stack installation: -l https://nadm50.dyn.nesc.nokia.net The fully qualified domain name (FQDN) is mandatory for dual-stack installation, however, you can also use FQDN in IPv4 and IPv6 installation modes.
-s, --south-ip	This parameter is mandatory, and required for the southbound address. This IP address is used to manage hardware devices.
-t, --snmp-ip	This parameter is mandatory and must be different from the southbound IP address (--south-ip). This is the address of the MantaRay HW Manager Simple Network Management Protocol (SNMP) trap listener.
-u, --auth-admin	This parameter defines the name for the admin user account. The admin user account will be created in the authentication system (Keycloak) during the installation. If you don't specify the parameter, the <i>admin</i> default value is used.
-p, --auth-pass	This parameter defines the password for the main user. If you don't specify it, a random password will be generated during the installation and displayed to the user. Note: Remember to introduce the password in single quotes.

Parameter name	Description
-e, --admin-email	This parameter defines an e-mail address of the admin user. If you don't specify the address, the <i>automatic@nokia.com</i> default value is used.
-w, --hw-manager-user	This parameter defines the initial user of MantaRay HW Manager. The MantaRay HW Manager user account for the MantaRay HW Manager northbound login is created during the installation. If you don't specify the parameter, the <i>autogenerated</i> default value is used.
-n, --hw-manager-pass	This parameter defines the password for the initial MantaRay HW Manager user. If you don't specify the parameter, a random password is generated during the installation and displayed.  Note: Remember to introduce the password in single quotes.
-U, --username	This parameter is mandatory for the user on the Kubernetes node. This parameter concerns the local user on each of the nodes and requires sudo permission.
-P, --password	This parameter defines the local user password on each of the nodes.  Note: Remember to introduce the password in single quotes.

Parameter name	Description
-K, --ssh-key	This parameter defines the path to the key to use for SSH key-based authentication. If not provided, password (-P) is used for authentication. You can test the ssh connectivity with the key using the following command : <pre>ssh <user>@<host> -i <key_file></pre> Example: <pre>ssh nuser@10.20.30.50 -i /home/nuser/.ssh/node.key</pre> Note: The path above is an example. The key can be stored anywhere on the host. Make sure you have the right file permission, at least 600: <pre>chmod 600 /home/<user>/.ssh/node.key</pre>
-s6, --south-ip6	This parameter defines the southbound IPv6 address. It is required for IPv6 or dual-stack installation. You need to provide the IPv6 address in a compressed format.
-t6, --snmp-ip6	This parameter defines the SNMP IPv6 address. It is required for IPv6 or dual-stack installation. You need to provide the IPv6 address in a compressed format.
-q, --sb-fqdn	This parameter defines the southbound FQDN. It adds FQDN to the southbound subject alternative name (SAN) to allow the registry usage with the name. It's required for installing Nokia Edge Automation Tool (NEAT) in IPv6 or dual stack mode. Don't use this parameter with standalone MantaRay HW Manager. Example: <pre>-q my.southbound.neat.net</pre>

Parameter name	Description
<code>-z, --size</code>	This parameter is optional and defines the planned MantaRay HW Manager installation size. The size refers to the estimated maximum amount of managed devices. If you provide this parameter, the <code>xxL</code> size value is valid. This is the value of an xx-large MantaRay HW Manager installation, which is up to 7000 managed devices. Specifying this parameter is needed to ensure that the <code>xxL</code> size works correctly. The specified size indicates a sufficient amount of host resources (for example, memory) that needs to be requested by the MantaRay HW Manager Kubernetes services. If the requested memory requirements aren't fulfilled in the host, for example, if the value is <code>-z xxL</code>, the installation is stopped.
<code>-h, --help</code>	This parameter displays the help message and the available options.

Run the `./initialize` script with the following parameters. The parameters presented in the example:

- create an admin user to the authentication system.
- create a password for the admin user.
- define an email address for the admin user.
- create a user for MantaRay HW Manager.
- create a password for the MantaRay HW Manager user.
- define the southbound IP address and the host URL where MantaRay HW Manager is running.

Note:

In the examples below, you need to use either the user SSH login password (`-P` parameter) or the user SSH key (`-K` parameter).

Step example

- IPv4-only installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://10.37.44.185 -s 192.168.1.43 \
-t 192.168.1.41 -u admin -p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' -U nuser -P 'Pa55w#rd'
```

With the user SSH key (the `-K` parameter):

```
cdcd nadcm-<release>/
sudo ./initialize -l https://10.37.44.185 -s 192.168.1.43 \
-t 192.168.1.41 -u admin -p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' -U nuser -K "home/nuser/.ssh/node.key"
```

■ IPv6-only installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://[2001:dcde:0:61::50] \
-s6 fd00:deed:acdc:1908::50 \
-t6 fd00:deed:acdc:1908::50 -u admin -p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' -U nuser -P 'Pa55w#rd'
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://[2001:dcde:0:61::50] \
-s6 fd00:deed:acdc:1908::50 \
-t6 fd00:deed:acdc:1908::50 -u admin -p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' -U nuser -K "home/nuser/.ssh/node.key"
```

Note:

Remember that the `-s6` and `-t6` addresses should be entered in a compressed format.

■ Dual-stack installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://nadcm50.dyn.nesc.nokia.net -s 192.168.8.50 \
-t 192.168.8.50 -s6 fd00:deed:acdc:1908::50 -t6 fd00:deed:acdc:1908::50 \
-u admin -p 'P@ssw0rd' -e automatic@nokia.com -w nadcmadmin -n 'P@55w0rd' \
-U nuser -P 'Pa55w#rd'
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize -l https://nadcm50.dyn.nesc.nokia.net -s 192.168.8.50 \
-t 192.168.8.50 -s6 fd00:deed:acdc:1908::50 -t6 fd00:deed:acdc:1908::50 \
-u admin -p 'P@ssw0rd' -e automatic@nokia.com -w nadcmadmin -n 'P@55w0rd' \
-U nuser -K "home/nuser/.ssh/node.key"
```

Note:

In the case of the dual-stack installation, remember that:

- ▣ the `-l` parameter needs to be an FQDN. For example:
https://nadcm.dyn.mydomain.net.
- ▣ the northbound and the southbound interfaces need to be dual-stack.
- ▣ the `-s6` and `-t6` addresses should be entered in a compressed format.

Step result

Expected results (the IP and password will be different according to the site):

```
***** BINARY INTEGRITY *****
```

Note! Execute binary integrity check again using script:

```
./scripts/binary_integrity_check.sh
```

Note! Release package binary integrity can be checked manually (without script 'binary_integrity_check.sh') by executing in command line:

```
sha256sum -c --quiet binary_integrity/release_files/nadcm*
```

(result shall notify from changed or missing files. New files are not noted.)

To setup authentication or to add new users:

```
https://10.21.20.252/auth/admin/
```

```
username: admin
```

```
password: P@ssw0rd
```

To login to MantaRay HW Manager as an autogenerated user:

```
https://10.21.20.252
```

```
username: nadcmadmin
```

```
password: P@55w0rd
```

NOTE: Please store all the credentials above carefully

Result

You have installed high available local MantaRay HW Manager. You can proceed to [Post-installation actions](#).

5.3 Installing central MantaRay HW Manager

You can install the central MantaRay HW Manager application only on the single-node infrastructure.

Procedure

- 1 Log in as non-root user.

Note:

Ensure that the non-root user can run commands with `sudo`. The same user account is needed on all three hosts, where MantaRay HW Manager is installed.

2 Start the central MantaRay HW Manager installation.

```
sudo ./initialize-central
```

Table 9: Parameters required for installing central MantaRay HW Manager with the single-node infrastructure

Parameter name	Description
<code>-l, --hosturl</code>	This parameter is mandatory, and required for the northbound URL. The host URL allows access to the MantaRay HW Manager GUI from a browser. Example of the <code>--hosturl</code> value: - For IPv4 installation: <code>-l https://10.37.44.185</code> - For IPv6 installation, you need to provide the IPv6 address in a compressed format: <code>-l https://[2001:dcde:0:61::50]</code> - For dual-stack installation: <code>-l https://nadm50.dyn.nesc.nokia.net</code> The fully qualified domain name (FQDN) is mandatory for dual-stack installation, however, you can also use it in IPv4 and IPv6 installation modes.
<code>-u, --auth-admin</code>	This parameter defines the name for the admin user account. The admin user account will be created in the authentication system (Keycloak) during the installation. If you don't specify the parameter, the <i>admin</i> default value is used.
<code>-p, --auth-pass</code>	This parameter defines the password for the main user. If you don't specify it, a random password will be generated during the installation and displayed to the user.  Note: Remember to introduce the password in single quotes.
<code>-e, --admin-email</code>	This parameter defines an e-mail address of the admin user. If you don't specify the address, the <i>automatic@nokia.com</i> default value is used.
<code>-w, --hw-manager-user</code>	This parameter defines the initial user of MantaRay HW Manager. The MantaRay HW Manager user account for the MantaRay HW Manager northbound login is created during the installation. If you don't specify the parameter, the <i>autogenerated</i> default value is used.

Parameter name	Description
-n, --hw-manager-pass	This parameter defines the password for the initial MantaRay HW Manager user. If you don't specify the parameter, a random password is generated during the installation and displayed.  Note: Remember to introduce the password in single quotes.
-U, --username	This parameter is mandatory for the user on the Kubernetes node. This parameter concerns the local user on each of the nodes and requires sudo permission.
-P, --password	This parameter defines the local user password on each of the nodes.  Note: Remember to introduce the password in single quotes.
-K, --ssh-key	This parameter defines the path to the key to use for SSH key-based authentication. If not provided, password (-P) is used for authentication. You can test ssh connectivity with the key using the following command : <pre>ssh <user>@<host> -i <key_file></pre> Example : <pre>ssh nuser@10.20.30.50 -i /home/user/.ssh/node.key</pre>  Note: The path above is an example. The key can be stored anywhere on the host. Make sure you have the right file permission, at least 600: <pre>chmod 600 /home/<user>/.ssh/node.key</pre>
-m, --install-mode	This parameter is mandatory and it defines the central MantaRay HW Manager installation mode. Possible values: • ipv4 • ipv6 • dual
-h, --help	This parameter shows the help message and the available options.

Run the initialization script with the following parameters. The parameters presented in the example:

- create an admin user to the authentication system.
- create a password for the admin user.
- define an email address for the admin user.
- create a user for MantaRay HW Manager.
- create a password for the MantaRay HW Manager user.
- define the southbound IP address and the host URL where MantaRay HW Manager is running.

Note:

In the examples below, you need to use either the user SSH login password (`-P` parameter) or the user SSH key (`-K` parameter).

Step example

- IPv4-only installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize-central -l https://172.16.2.178 -u admin \
-p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' \
-U nuser -P 'Pa55w#rd' \
-m ipv4
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize-central -l https://172.16.2.178 -u admin \
-p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' \
-U nuser -K "home/nuser/.ssh/node.key" \
-m ipv4
```

- IPv6-only installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize-central -l https://[2001:dcdc:0:61::50] -u admin \
-p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' -U nuser -P 'Pa55w#rd' \
-m ipv6
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize-central -l https://[2001:dcdc:0:61::50] -u admin \
-p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' -U nuser -K "home/nuser/.ssh/node.key" \
-m ipv6
```

■ Dual-stack installation

With the user SSH login password (the `-P` parameter):

```
cd nadcm-<release>/
sudo ./initialize-central -l https://nadcm-central.abc.net -u admin \
-p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' -U nuser -P 'Pa55w#rd' \
-m dual
```

With the user SSH key (the `-K` parameter):

```
cd nadcm-<release>/
sudo ./initialize-central -l https://nadcm-central.abc.net -u admin \
-p 'P@ssw0rd' -e automatic@nokia.com \
-w nadcmadmin -n 'P@55w0rd' -U nuser -K "home/nuser/.ssh/node.key" \
-m dual
```

Note:

In the case of the dual-stack installation, remember that:

- the `-l` parameter needs to be an FQDN. For example:
https://nadcm.dyn.mydomain.net.
- the northbound and the southbound interfaces need to be dual-stack.

Step result

Expected results (the IP and password will be different according to the site):

```

***** BINARY INTEGRITY *****
Note! Execute binary integrity check again using script:
    ./scripts/binary_integrity_check.sh

Note! Release package binary integrity can be checked manually (without
script 'binary_integrity_check.sh') by executing in command line:
    sha256sum -c --quiet binary_integrity/release_files/nadcm*
    (result shall notify from changed or missing files. New files are not
    noted.)

To setup authentication or to add new users:
    https://10.21.20.252/auth/admin/
    username: admin
    password: P@ssw0rd

To login to MantaRay HW Manager as an autogenerated user:
    https://10.21.20.252
    username: nadcmadmin
    password: P@55w0rd

NOTE: Please store all the credentials above carefully

```

Result

You have installed the central MantaRay HW Manager application. You can now proceed to [Post -installation actions](#).

5.4 Installing the MantaRay HW Manager on OpenShift Container Platform

The procedure for local (SNO/MNO) and central MantaRay HW Manager application installation on OpenShift Container Platform (OCP).

Before you start

See the [Prerequisites for installing the MantaRay HW Manager on OCP](#).

Note:

MantaRay HW Manager installation on OCP must be initiated from the installer (provisioner) node used to deploy the OCP cluster. The installer must have network connectivity to both the OCP cluster API and the container registry endpoints.

Procedure

- 1 Log in as a non-root user.

Note:

Use the `sudo -l` command to ensure that the non-root user's local account can perform the steps below with `sudo` privileges.

- 2 Prepare the installation package.

Download the MantaRay HW Manager installation package to the provisioner/installer node, preferably under `/data/hwm/`.

- 3 **[Optional]** Verify the integrity of the downloaded package.

```
# cd /data/hwm/
# sha256sum -c nm-ocp-software-OCPCS_VERSION_TAG.iso.sha2.txt
```

- 4 Extract the release

```
# tar zxvf nadcm_full-<release>.tgz
```

- 5 Set the environment variables

Mandatory variables:

- Path to kubeconfig file:

```
# export KUBECONFIG=<path_to_kubeconfig_file>
```

Example:

```
# export KUBECONFIG=/home/admin/kubeconfig_142.yaml
```

- Container manager:

```
# export ContainerManager=podman
```

- Storage class name (get from 'kubectl get storageclass'):

```
# export OCPSTORAGECLASS=<storage_class_name>
```

Example:

```
# export OCPSTORAGECLASS=lvms-vg1
```

- OCP internal or external registry route:

```
# export IMAGEREGISTRY_ADDRESS=default-route-openshift-image-registry.apps.<cluster_name>.<base_domain>
```

Example:

```
# export IMAGEREGISTRY_ADDRESS=default-route-openshift-image-registry.apps.cluster01.example.com
```

- Image repository name:

```
# export IMAGEREGISTRY_REPOSITORY=hwm
```

- Cluster admin username with push/pull permissions:

```
# export IMAGEREGISTRY_USERNAME=<cluster-admin username>
```

Example:

```
# export IMAGEREGISTRY_USERNAME=admin
```

- Token from oc login command:

```
# export IMAGEREGISTRY_PASSWORD=<token got from oc login command>
```

- Skip ping test to cluster nodes during installation:

```
# export IGNORE_PING=true
```

- Set the FQDN environmental variable value:

```
# export FQDN=<host URL>
```

Optional variables:

- Temporary folder for HWM images (default: /tmp):

```
# export IMGTMPDIR=/data/hwm/images
```

6 Start the MantaRay HW Manager installation

See [Installing the MantaRay HW Manager application](#) for detailed arguments description.

- Example command for local MantaRay HW Manager installation for single/multi-node OCP/NMCCP with both IPv4 and IPv6:

```
# sudo -E ./initialize -l
https://nadcm.apps.mrhwm-sno-2.lab.neat.nsn-rdnet.net -s 192.0.2.1 -s6
2001:0db8:0000:0000:0000:0000:0000:0001 -t 192.0.2.2 -t6
2001:0db8:0000:0000:0000:0000:0000:0002 -u admin -p 'P@ssw0rd' -e
automatic@nokia.com -w nadcmadmin -n 'P@ssw0rd' -U core
```

Expected result:

```
***** BINARY INTEGRITY *****
Note! Execute binary integrity check again using script:
    sudo ./scripts/binary_integrity_check.sh
***WARNING***
Please note that error(s) has been found in binary integrity check.
To setup authentication or to add new users:
    https://nadcm.apps.mrhwm-sno-2.lab.neat.nsn-rdnet.net/auth/admin/
    username: admin
    password: P@ssw0rd
To login to MantaRay HW Manager:
    https://nadcm.apps.mrhwm-sno-2.lab.neat.nsn-rdnet.net
    username: nadcmadmin
    password: P@55w0rd
NOTE: Please store all the credentials above carefully
```

- Example command for central MantaRay HW Manager installation on single node OCP:

```
# sudo -E ./initialize-central -l
https://nadcmm.apps.mrhwm-sno-1.lab.neat.nsn-rdnet.net -u admin -p
'P@ssw0rd' -e authorization.admin@foobar.org -w nadcmmadmin -n 'P@ssw0rd'
-U core -m ipv4
```

Expected result:

```
***** BINARY INTEGRITY *****
Note! Execute binary integrity check again using script:
    sudo ./scripts/binary_integrity_check.sh
***WARNING***
Please note that error(s) has been found in binary integrity check.
To setup authentication or to add new users:
    https://nadcmm.apps.mrhwm-sno-2.lab.neat.nsn-rdnet.net/auth/admin/
    username: admin
    password: P@ssw0rd
To login to MantaRay HW Manager:
    https://nadcmm.apps.mrhwm-sno-2.lab.neat.nsn-rdnet.net
    username: nadcmmadmin
    password: P@55w0rd
NOTE: Please store all the credentials above carefully
```

6. Post-installation actions

You need to perform the post-installation actions to log in to MantaRay HW Manager, create a first user, and log in to Keycloak.

Before you start

Note:

For details on integrating with MantaRay, see *Integrating MantaRay HW Manager with MantaRay NM*.

Note:

You can perform this procedure if you log in as a non-root user. Ensure that the non-root user can run commands with `sudo`. The only exception is [step 3](#) with its substeps, for which you need root user access.

Procedure

- 1 Write down the credentials.

For central MantaRay HW Manager, you can check the credentials from the `central-values.yaml` configuration.

For local MantaRay HW Manager, after the installation has finished, the script outputs the user management system credentials and the autogenerated MantaRay HW Manager user credentials. You need to record those credentials to access the MantaRay HW Manager application and add new users in Keycloak.

Example of the script output:

```
run 'export KUBECONFIG=/root/admin.conf' in order to use kubectl
```

To setup authentication or to add new users:

```
https://sut4-server1-pub.sut-4.archcommon.nsn-rdnet.net/auth/admin/
```

```
username: admin
```

```
password: P@ssw0rd
```

To login to AirFrame Data Center Manager as an autogenerated user:

```
https://sut4-server1-pub.sut-4.archcommon.nsn-rdnet.net
```

```
username: nadcmadmin
```

```
password: P@55w0rd
```

NOTE: Please store all the credentials above carefully

2 Wait for Kubernetes to set up services.

After the installation script has finished, it may take several minutes before the MantaRay HW Manager system is accessible through the northbound address.

3 Configure the host system cron job to get the Kubernetes internal service certificates renewal done automatically.

Note:

You need to have the Kubernetes version 1.21.1 or newer. Before continuing, please check your Kubernetes version by logging into the MantaRay HW Manager host operating system (OS) and running the `kubeadm version` command. You need to use root user access to perform this step and its substeps.

3.1 Create the script on the MantaRay HW Manager host.

```
cat <<EOF > /etc/cron.monthly/kube-certs-renewal.sh
#!/bin/sh

# Renewal of all kubernetes certificates
/usr/local/bin/kubeadm certs renew all

EXITVALUE=$?
if [ \${EXITVALUE} != 0 ]; then
    /usr/bin/logger -t kube-certs-renewal "ALERT exited abnormally
with [\${EXITVALUE}]"
else
    yes | cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
fi
exit 0
EOF
```

Make the script executable:

```
chmod 755 /etc/cron.monthly/kube-certs-renewal.sh
```

Note:

Nokia recommends running the script manually once to see that it also works with cron. You can run the following cron job script:

```
/etc/cron.monthly/./kube-certs-renewal.sh
```

Make sure that MantaRay HW Manager host system cron job is functioning as expected.

3.2 [Optional] Renew the Kubernetes certificate:

See [Renewing valid Kubernetes certificates](#) for more details.

Note:

You can renew the Kubernetes certificates at any time.

4 Check the validity of the certificates.

```
kubeadm certs check-expiration
```

Step example

Figure 7: Example output of the certificate validity check

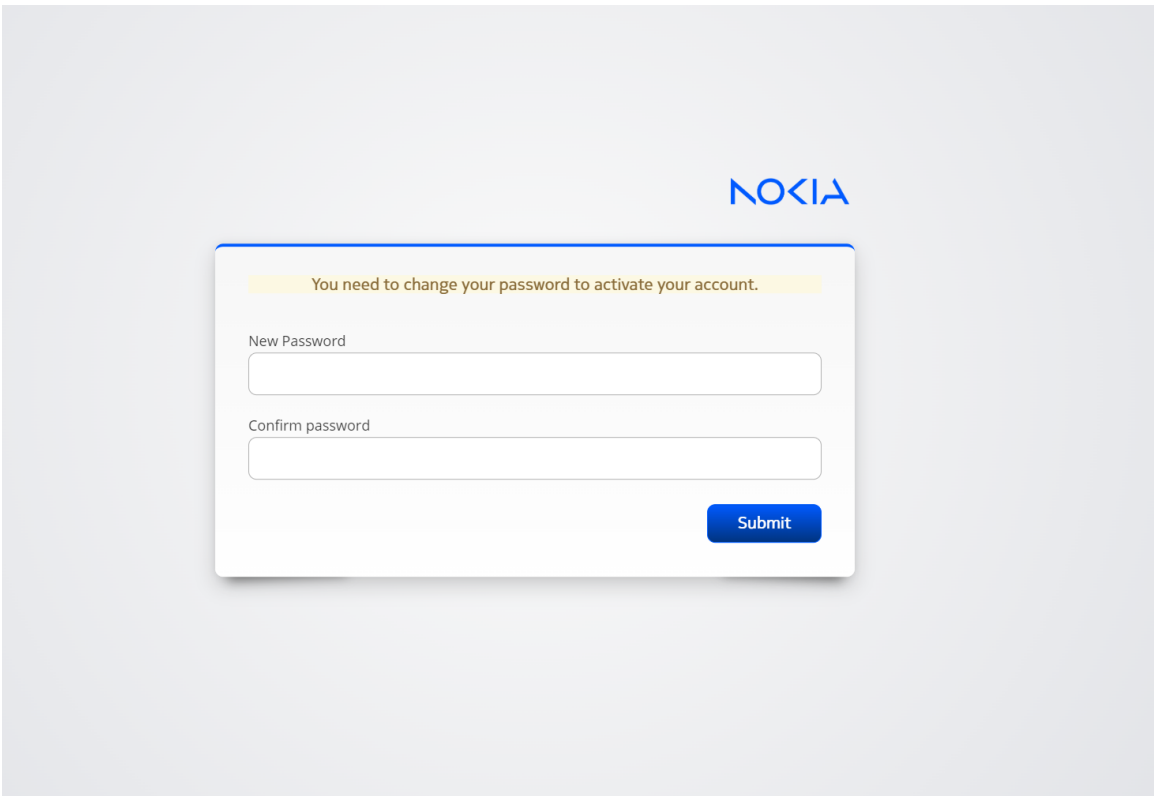
CERTIFICATE	EXPIRES	RESIDUAL TIME	CERTIFICATE AUTHORITY	EXTERNALLY MANAGED
admin.conf	Sep 29, 2023 06:39 UTC	364d		no
apiserver	Sep 29, 2023 06:39 UTC	364d	ca	no
apiserver-kubelet-client	Sep 29, 2023 06:39 UTC	364d	ca	no
controller-manager.conf	Sep 29, 2023 06:39 UTC	364d		no
front-proxy-client	Sep 29, 2023 06:39 UTC	364d	front-proxy-ca	no
scheduler.conf	Sep 29, 2023 06:39 UTC	364d		no

CERTIFICATE AUTHORITY	EXPIRES	RESIDUAL TIME	EXTERNALLY MANAGED
ca	Dec 12, 2031 14:53 UTC	9y	no
front-proxy-ca	Dec 12, 2031 14:53 UTC	9y	no

5 Log in to MantaRay HW Manager and change the password.

When you log in to the system for the first time, you need to change the password.

Figure 8: View of the first login to MantaRay HW Manager



The complete installation package isn't needed for MantaRay HW Manager operations, but it contains some scripts (such as the log collecting script) which you may need in the future. Therefore, don't delete the latest installation directory.

6 Verify the pod and the deployment status.

```
sudo kubectl -n nadcm get pods
sudo kubectl -n nadcm get deployments
```

Result

MantaRay HW Manager installation is now ready and you can continue by creating a new user.

6.1 IP forwarding

You need to make sure that IP forwarding is enabled in your host operating system (OS) in the `kernel tunables`.

Purpose

Note:

IP forwarding chapter is not applicable for installation on OpenShift Container Platform (OCP).

The IP forwarding enables the forwarding of traffic between network interfaces. The MantaRay HW Manager container environment requires continuously enabled IP forwarding in the host OS.

The IP forwarding is enabled based on the installation mode:

- IPv4-only: IPv4 forwarding is enabled
- Dual-stack: Both IPv4 and IPv6 forwarding are enabled
- IPv6-only: IPv6 forwarding is enabled

Note:

If the IP forwarding is disabled in the *kernel tunables* or gets disabled at runtime, the MantaRay HW Manager containers can't communicate, so the application stops working upon the next host OS reboot.

Procedure

- 1 Check the IP forwarding settings in the *kernel tunables*.

For IPv4:

```
[root@user admin] # sysctl -a | grep "net.ipv4.ip_forward"
```

For IPv6:

```
[root@user admin] # sysctl -a | grep "net.ipv6.conf.all.forwarding"
```

If the `net.ipv4.ip_forward` and `net.ipv6.conf.all.forwarding` parameters are `1`, it means the IPv4 and IPv6 addresses are enabled. Otherwise, go to step 2.

- 2 Open the `/etc/sysctl.d/999-k8s.vanilla.conf` file using the command-line editor.

Tip:

You can edit the parameter using the `vi` command-line editor.

Note:

By default, Linux distributions don't explicitly disable forwarding at `/etc/sysctl.conf`. If disabling is present, there may be a conflict concerning the server purpose. Before installing MantaRay HW Manager, you need to remove the disabling lines or, optionally, set them to enable forwarding.

- 3 Enable IP forwarding.

Step result

```
net.ipv4.ip_forward=1
```

```
net.ipv6.conf.all.forwarding=1
```

- 4 Ensure that no other installed and running software in the MantaRay HW Manager host OS doesn't disable the IP forwarding.

! Notice:

After the installation process is finished, MantaRay HW Manager can't influence the IP forwarding setting.

You need to make sure that any other software running in the host OS alongside the MantaRay HW Manager application doesn't disable IP forwarding by setting the `net.ipv4.ip_forward` and `net.ipv6.conf.all.forwarding` parameters to 0.

Result

You have enabled the IP forwarding.

6.2 Configuring log forwarding

You can configure log forwarding to forward the Kubernetes and MantaRay HW Manager logs to the remote rocket-fast system for log processing (rsyslogd).

Purpose

i Note:

Configuring log forwarding chapter is not applicable for installation on OpenShift Container Platform (OCP).

The MantaRay HW Manager application logs are, by default, using *kubectl logs* commands as *.log* files on each cluster node in the `var/log/container` files.

If you want to forward the Kubernetes or MantaRay HW Manager logs to a central log system, you need to follow an optional procedure using the *fluentd* collector.

Fluentd is an open-source log collector that you can use to aggregate the logs from each node and push them to a remote server.

Before you start

Make sure you have:

- a proper *rsyslog* server configured to receive remote logs.
- written down the following *rsyslog* server data:
 - IP address
 - Port
 - Protocol - User Datagram Protocol (UDP) and Transmission Control Protocol (TCP)

Procedure

- 1 In the MantaRay HW Manager application packet, go to `nadcm-helm -charts/misc/fluentd`.

- 2 Update the *fluent.yaml* file.

Set the following *ENV* values:

- `SYSLOG_HOSTNAME`: it identifies the source of the logs in *syslog* server, uniquely identifies the MantaRay HW Manager instance. The default value is *nadcm*.
- `SYSLOG_HOST`: the IP address of the remote *syslog* server.
- `SYSLOG_PROTOCOL`: UDP or TCP.

Check the *fluent.conf* file for other possible *ENV* values such as `PORT` or `SSL`.

For more information about the *fluentd* configuration, see [↗ Introduction - Fluentd](#).

- 3 Create a ConfigMap.

```
kubectl create configmap -n kube-system fluentd-nadcm --from
-file=kubernetes.conf --from-file=fluent.conf
```

- 4 Install the Kubernetes resources.

```
kubectl apply -f fluent.yaml
```

- 5 Check if the *fluentd* pods are running.

```
kubectl get pods -n kube-system | grep fluentd
```

Result

You have configured the log forwarding.

Tip:

You can configure the rsyslog server to match the hostname set as an *ENV* value, using the following line:

```
if $hostname == 'nadcm' then /var/log/nadcm.log
```

For the *ENV* value reference, see [step 2. Update the fluent.yaml file](#).

You can also filter the MantaRay HW Manager logs only with the following command:

```
tail -f /var/log/nadcm.log | grep '"namespace":"nadcm"'
```

Example

Log entry:

```
{
  "message": "2023-05-09T09:24:38.127498224+03:00 stdout F INFO: 2023/05/09
06:24:38 :mongo.go:70 Mongo native client created",
  "node": "node3",
  "namespace": "nadcm",
  "pod": "omagent-69c59588cb-t5wd2",
  "container": "sftp",
  "container_id":
"60259689b40d5f9c4ad4c192dc9d420d0757e7126eb6a6d656a454293a991b08"
}
```

6.2.1 Disabling log forwarding

You can disable log forwarding using `kubectl` commands.

Purpose

You can stop forwarding the Kubernetes and MantaRay HW Manager logs by disabling `fluentd`.

Procedure

- 1 [Optional] Back up the current DaemonSet.

```
kubectl get ds/fluentd -n kube-system -o yaml > fluentd_ds.yaml
```

- 2 Remove the DaemonSet.

```
kubectl delete ds/fluentd -n kube-system
```

Result

You have disabled the log forwarding.

Note:

After disabling `fluentd`, you can recreate the DaemonSet using a backup file. To do so, you need to first edit the `fluent_ds.yaml` file to remove some of the properties generated by Kubernetes. After editing the file, you can run the following command:

```
kubectl apply -f fluentd_ds.yaml
```

6.3 Configuring login message in MantaRay HW Manager

You can configure the login message of MantaRay HW Manager using a ConfigMap.

Purpose

You can find a customizable MantaRay HW Manager login message below the user credentials fields.

Figure 9: MantaRay HW Manager login message

Note:

The custom login message policy:

- Create a text that contains maximum 1024 characters.
- Don't use spaces and line breaks at the beginning and at the end of the message.
- Don't use the following characters: < > \$! ? & % @ #.

Procedure

1 To create a customized login message, perform the following steps.

1.1 Create a *keycloak-theme* ConfigMap.

You can create the *keycloak-theme* ConfigMap in two ways:

- Create the ConfigMap inserting the login message text as a literal value:

```
kubectl -n nadcm create configmap keycloak-theme --from
-literal=theme=<theme text>
```

i Note:

You need to name the key as *theme*.

Step example

```
kubectl -n nadcm create configmap keycloak-theme --from
-literal=theme="NADCM is in maintenance now. Avoid login"
```

- Create the ConfigMap from a file that contains the login message text:

```
kubectl -n nadcm create configmap keycloak-theme --from-file=./theme
```

i Note:

You need to name the file as *theme*.

1.2 [Optional] Edit the existing theme using the `kubectl replace -f` command.

Step example

- Editing the text in the ConfigMap from the literal value:

```
kubectl -n nadcm create configmap keycloak-theme --from
-literal=theme="HELLO WORLD" -o yaml --dry-run=client | kubectl
replace -f -
```

- Editing the text in the ConfigMap from the file:

```
kubectl -n nadcm create configmap keycloak-theme --from
-file=./theme -o yaml --dry-run=client | kubectl replace -f
```

i Note:

Keycloak restarts automatically when the *keycloak-theme* ConfigMap is created or changed.

The Keycloak restart takes up to three minutes.

Step result

You have created the new MantaRay HW Manager login message.

2 [Optional] To return to the default theme, perform the following steps.

2.1 Delete the ConfigMap using the following command:

```
kubectl -n nadcm delete configmap keycloak-theme
```

2.2 Restart Keycloak using the following command:

```
kubectl -n nadcm delete pod keycloak-0 keycloak-1 keycloak-2
```

Step result

You have returned to the default MantaRay HW Manager login message.

7. Renewing Kubernetes certificates

If Kubernetes certificates expire, you can't perform any life cycle management (LCM) actions for the Kubernetes pods. You can renew the certificates manually or automatically.

Note:

Renewing Kubernetes certificates chapter is not applicable for installation on OpenShift Container Platform (OCP).

Kubernetes sets a one-year validity period for its internal service certificates. If the certificates expire, the application running in the cluster works, but you can't perform any LCM actions for the pods managed by Kubernetes. For example, Kubernetes can't restart pods after the pod liveness probe failure or after the MantaRay HW Manager host reboot.

You can renew the Kubernetes certificates at any time, whether or not they have expired.

There are two ways of renewing certificates:

- automatically, by configuring the host system cron job. For more information, see steps 3 and 4 in [Post-installation actions](#).

Note:

Nokia recommends the automatic way of renewing the Kubernetes certificates.

- manually. For more information, see [Renewing Kubernetes certificates manually](#).

7.1 Checking the Kubernetes certificates validity

You need to check the validity of the certificates before running the renewing command, as the post action differs depending on whether the certificates are expired or valid.

Before you start

Note:

Ensure that the non-root user can run commands with `sudo`.

Procedure

- 1 Check the validity of the certificates.

```
sudo kubeadm certs check-expiration
```

Step result

- If the *RESIDUAL TIME* column indicates the *<invalid>* status, the certificates are expired. For further instructions, go to [Renewing expired Kubernetes certificates](#).

```
[check-expiration] Reading configuration from the cluster...
[check-expiration] FYI: You can look at this config file with 'kubectl -n
kube-system get cm kubeadm-config -o yaml'
[check-expiration] Error reading configuration from the Cluster. Falling
back to default configuration
```

CERTIFICATE	EXPIRES	RESIDUAL
TIME	CERTIFICATE AUTHORITY	EXTERNALLY MANAGED
admin.conf	Jan 18, 2024 10:00 UTC	<invalid>
ca	no	
apiserver	Jan 18, 2024 10:00 UTC	<invalid>
ca	no	
!MISSING! apiserver-etcd-client		
apiserver-kubelet-client	Jan 18, 2024 10:00 UTC	<invalid>
ca	no	
controller-manager.conf	Jan 18, 2024 10:00 UTC	<invalid>
ca	no	
!MISSING! etcd-healthcheck-client		
!MISSING! etcd-peer		
!MISSING! etcd-server		
front-proxy-client	Jan 18, 2024 10:00 UTC	<invalid>
front-proxy-ca	no	
scheduler.conf	Jan 18, 2024 10:00 UTC	<invalid>
ca	no	

- If the *RESIDUAL TIME* column indicates time, the certificates are valid. For further instructions, go to [Renewing valid Kubernetes certificates](#).

```
[check-expiration] Reading configuration from the cluster...
[check-expiration] FYI: You can look at this config file with 'kubectl -n
kube-system get cm kubeadm-config -o yaml'
```

CERTIFICATE	EXPIRES	RESIDUAL TIME
CERTIFICATE AUTHORITY	EXTERNALLY MANAGED	
admin.conf	Mar 11, 2025 16:43 UTC	362d ca
no		
apiserver	Mar 11, 2025 16:43 UTC	362d ca
no		
apiserver-kubelet-client	Mar 11, 2025 16:43 UTC	362d ca
no		
controller-manager.conf	Mar 11, 2025 16:43 UTC	362d ca
no		
front-proxy-client	Mar 11, 2025 16:43 UTC	362d
front-proxy-ca	no	
scheduler.conf	Mar 11, 2025 16:43 UTC	362d ca
no		

CERTIFICATE AUTHORITY	EXPIRES	RESIDUAL TIME
EXTERNALLY MANAGED		
ca	Mar 09, 2034 16:43 UTC	9y no
front-proxy-ca	Mar 09, 2034 16:43 UTC	9y no

7.2 Renewing Kubernetes certificates manually

You can renew Kubernetes certificates manually. Manual renewal procedures depend on whether the certificates have expired or not.

7.2.1 Renewing valid Kubernetes certificates

You can renew the valid Kubernetes certificates manually, using the `kubeadm cert renew all` command.

Purpose

If you renew certificates that are still valid, the renewing procedure doesn't cause any service breaks on the application side.

Note:

All presented outcome cases may vary as they depend on the version of Kubernetes.

Before you start

Make sure that:

- the non-root user can run commands with `sudo`.
- Kubernetes certificates are valid. For more information, see [Checking the Kubernetes certificates validity](#).

Procedure

- 1 Renew the certificates.

```
sudo kubeadm certs renew all
```

Step result

```
[renew] Reading configuration from the cluster...
[renew] FYI: You can look at this config file with 'kubectl -n kube-system
get cm kubeadm-config -o yaml'

certificate embedded in the kubeconfig file for the admin to use and for
kubeadm itself renewed
certificate for serving the Kubernetes API renewed
certificate for the API server to connect to kubelet renewed
certificate embedded in the kubeconfig file for the controller manager to
use renewed
certificate for the front proxy client renewed
certificate embedded in the kubeconfig file for the scheduler manager to use
renewed

Done renewing certificates. You must restart the kube-apiserver, kube
-controller-manager, kube-scheduler and etcd, so that they can use the new
certificates.
```

Note:

The last line of the command indicates the Kubernetes pods you need to restart.

2 Copy the admin.conf file into kube/config.

You need to perform this step to ensure that access to Kubernetes is authorized.

Step example

```
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
```

3 Restart the Kubernetes control plane containers.

After successfully renewing the Kubernetes certificates, you need to restart the following control plane containers to ensure the new certificates are picked up by the running services. (Kubernetes will automatically restart these static pods after they are stopped):

- kube-apiserver
- kube-controller-manager
- kube-scheduler

```
sudo crictl pods --namespace kube-system --name 'kube-scheduler-*|kube-
-controller-manager-*|kube-apiserver-*' -q | xargs sudo crictl rmp -f
```

Step result

```
stopped sandbox <kube-scheduler POD ID>
stopped sandbox <kube-controller-manager POD ID>
stopped sandbox <kube-apiserver POD ID>
removed sandbox <kube-scheduler POD ID>
removed sandbox <kube-controller-manager POD ID>
removed sandbox <kube-apiserver POD ID>
```

4 Check if all Kubernetes control plane containers have restarted

```
sudo crictl pods --namespace kube-system --name 'kube-scheduler-*|kube-
-controller-manager-*|kube-apiserver-*'
```

Step result

POD ID	CREATED	STATE	NAME
NAMESPACE	ATTEMPT	RUNTIME	
ed0f8959b127c	1 second ago	Ready	kube-apiserver
-node1	kube-system	4	(default)
b4defd1bccd4d	1 second ago	Ready	kube-scheduler
-node1	kube-system	4	(default)
755a45788aac9	1 second ago	Ready	kube-controller
-manager-node1	kube-system	0	(default)

Verify whether containers are in the *Ready STATE*. The *CREATED* column corresponds to the container recreation time.

5 Restart the etcd service

```
sudo systemctl restart etcd.service
```

 Tip:

There is no outcome after the successful execution of this command.

Check the status of the etcd service by using the: `sudo systemctl --no-pager status etcd.service` command. The status of the successfully restarted service is: Active: active (running).

6 Check if all Kubernetes control plane pods are running.

```
sudo kubectl -n kube-system get pods
```

Step result

NAME	READY	STATUS	RESTARTS	AGE
calico-kube-controllers-648dff99-g4pqz 2d21h		1/1	Running	0
calico-node-nw9dz 2d21h		1/1	Running	0
coredns-77f7cc69db-s6sc4 2d21h		1/1	Running	0
dns-autoscaler-595558c478-m2q4d 2d21h		1/1	Running	0
kube-apiserver-node1 41m		1/1	Running	0
kube-controller-manager-node1 41m		1/1	Running	1
kube-proxy-slsvf 2d21h		1/1	Running	0
kube-scheduler-node1 41m		1/1	Running	1
local-volume-provisioner-vr8tj 2d21h		1/1	Running	0
metrics-server-bd6df7764-96dtn 2d21h		1/1	Running	0
nodelocaldns-rszc7 2d21h		1/1	Running	0

Verify whether all pods are in the *Running STATUS*.

Result

You have renewed valid Kubernetes certificates.

7.2.2 Renewing expired Kubernetes certificates

You can renew the expired Kubernetes certificates manually. During the renewal procedure, you need to restart kubelet and container runtime.

Purpose

If you renew expired Kubernetes certificates, expect a few minutes of service interruption.

Note:

All presented outcome cases may vary as they depend on the version of Kubernetes.

Before you start

Make sure that:

- the Kubernetes certificates are expired. For more information, see [Checking the Kubernetes certificates validity](#).
- the non-root user can run commands with `sudo`.

Procedure

- 1 Renew the certificates.

```
sudo kubeadm certs renew all
```

Step result

```
[renew] Reading configuration from the cluster...
[renew] FYI: You can look at this config file with 'kubectl -n kube-system
get cm kubeadm-config -o yaml'

certificate embedded in the kubeconfig file for the admin to use and for
kubeadm itself renewed
certificate for serving the Kubernetes API renewed
certificate for the API server to connect to kubelet renewed
certificate embedded in the kubeconfig file for the controller manager to
use renewed
certificate for the front proxy client renewed
certificate embedded in the kubeconfig file for the scheduler manager to use
renewed

Done renewing certificates. You must restart the kube-apiserver, kube
-controller-manager, kube-scheduler and etcd, so that they can use the new
certificates.
```

Note:

The last line of the command indicates the Kubernetes pods you need to restart.

2 Copy the admin.conf file into kube/config.

You need to perform this step to ensure that access to Kubernetes is authorized.

Step example

```
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
```

3 Restart the Kubernetes control plane containers.

After successfully renewing the Kubernetes certificates, you need to restart the following control plane containers to ensure the new certificates are picked up by the running services. (Kubernetes will automatically restart these static pods after they are stopped):

- kube-apiserver
- kube-controller-manager
- kube-scheduler

```
sudo crictl pods --namespace kube-system --name 'kube-scheduler-*|kube-
-controller-manager-*|kube-apiserver-*' -q | xargs sudo crictl rmp -f
```

Step result

```
stopped sandbox <kube-scheduler POD ID>
stopped sandbox <kube-controller-manager POD ID>
stopped sandbox <kube-apiserver POD ID>
removed sandbox <kube-scheduler POD ID>
removed sandbox <kube-controller-manager POD ID>
removed sandbox <kube-apiserver POD ID>
```

4 Check if all Kubernetes control plane containers have restarted

```
sudo crictl pods --namespace kube-system --name 'kube-scheduler-*|kube-
-controller-manager-*|kube-apiserver-*'
```

Step result

POD ID	CREATED	STATE	NAME
NAMESPACE	ATTEMPT	RUNTIME	
ed0f8959b127c	1 second ago	Ready	kube-apiserver
-node1	kube-system	4	(default)
b4defd1bccd4d	1 second ago	Ready	kube-scheduler
-node1	kube-system	4	(default)
755a45788aac9	1 second ago	Ready	kube-controller
-manager-node1	kube-system	0	(default)

Verify whether containers are in the *Ready STATE*. The *CREATED* column corresponds to the container recreation time.

5 Restart the etcd service

```
sudo systemctl restart etcd.service
```

💡 Tip:

There is no outcome after the successful execution of this command.
Check the status of the etcd service by using the: `sudo systemctl --no-pager status etcd.service` command. The status of the successfully restarted service is: Active: active (running).

6 Restart kubelet.**6.1 Run the restarting command:**

```
sudo systemctl restart kubelet.service
```

6.2 Check the status of kubelet.

```
sudo systemctl --no-pager status kubelet.service
```

Step result

The status information of the successfully restarted kubelet is Active: active (running).

7 Restart Docker.

```
sudo systemctl restart docker.service
```

💡 Tip:

If you get the following error during Docker restarting:

```
Failed to restart docker.service: Unit docker.service not found.
```

it means you have a newer version of the infrastructure, and the container runtime in use is containerd.io. In this case, go to [step 8. \[Optional\] \[Only for the containerd.io container runtime\] Restart containerd.](#)

8 [Optional] [Only for the containerd.io container runtime] Restart containerd.

8.1 Run the restarting command:

```
sudo systemctl restart containerd.service
```

8.2 Check the status of containerd.

```
sudo systemctl --no-pager status containerd.service
```

Step result

The status information of the successfully restarted containerd is **Active: active (running)**.

Note:

Restarting kubelet and container runtime (Docker or containerd.io) may restart all application pods. This procedure may last up to a few minutes. After all pods are in the *Running* status, the application works normally.

9 Check if all Kubernetes control plane pods are running.

```
sudo kubectl -n kube-system get pods
```

Step result

NAME	READY	STATUS	RESTARTS	AGE
calico-kube-controllers-648dff99-g4pqz		1/1	Running	0
2d21h				
calico-node-nw9dz		1/1	Running	0
2d21h				
coredns-77f7cc69db-s6sc4		1/1	Running	0
2d21h				
dns-autoscaler-595558c478-m2q4d		1/1	Running	0
2d21h				
kube-apiserver-node1		1/1	Running	0
41m				
kube-controller-manager-node1		1/1	Running	1
41m				
kube-proxy-slsvf		1/1	Running	0
2d21h				
kube-scheduler-node1		1/1	Running	1
41m				
local-volume-provisioner-vr8tj		1/1	Running	0
2d21h				
metrics-server-bd6df7764-96dtn		1/1	Running	0
2d21h				
nodelocaldns-rszc7		1/1	Running	0
2d21h				

Verify whether all pods are in the *Running* STATUS.

Result

You have renewed the expired Kubernetes certificates.

8. Upgrading MantaRay HW Manager

You can upgrade the MantaRay HW Manager application as a user with the privilege to run `sudo` commands.

Purpose

You can upgrade only the MantaRay HW Manager application. The upgrading procedure is the same for all three variants of MantaRay HW Manager:

- Single-node local MantaRay HW Manager
- High availability local MantaRay HW Manager
- Central MantaRay HW Manager

To update the infrastructure components, you need to re-install the infrastructure.

Note:

You can't change the MantaRay HW Manager IP installation mode during the upgrade.

To change the IP installation mode (to IPv6 or dual-stack), you need to remove the MantaRay HW Manager application and the infrastructure components, then follow the installation procedure for the chosen IP installation mode.

For more information, see the following chapters:

- [Removing MantaRay HW Manager and infrastructure](#)
- [Installing infrastructure](#)
- [Installing the MantaRay HW Manager application](#)

Before you start

Make sure that:

- you have an up-to-date backup of the MantaRay HW Manager data.
For more information, see the *Backup of MantaRay HW Manager data* chapter in the *Operating and Maintaining MantaRay HW Manager* document.
- the current MantaRay HW Manager installation is running correctly.
To verify the installation, run the following command from the MantaRay HW Manager host operating system (OS) command line:

```
kubectl get pods -A
```

If the MantaRay HW Manager installation is running correctly, all MantaRay HW Manager pods have the *Running* status.

Notice:

If some of the MantaRay HW Manager pods are in a state other than *Running*, please contact Nokia Care before proceeding with the upgrade.

Tip:

In some cases, the MantaRay HW Manager host maintenance activities (for example, reboot) lead to creating parallel instances of the *Running* pods. Such parallel pods are in other states than *Running*, such as *Error* or *Completed*, and are terminated. If MantaRay HW Manager is functioning correctly, you can delete the terminated pods by running the `remove-parallel-pods.sh` script located in `<nadcm_installation_directory>/scripts`:

```
./remove-parallel-pods.sh
```

To automate the removal of parallel pods after each host reboot, Nokia recommends adding the script execution to `crontab`. To do so, open a crontab editor:

```
crontab -e
```

and add the following line:

```
@reboot sleep 900 &&  
<absolute_path_to_nadcm_installation_dir>/scripts/remove-parallel  
-pods.sh
```

As a result, cron executes the script 15 minutes after each host reboot.

Procedure

- 1 Download MantaRay HW Manager from Software Supply Tool (SWST) on the [Support Portal](#).

2 Verify the integrity of the downloaded package.

```
sha256sum -c <checksum file> (e.g. sha256sum -c nadcm_full-  
-<release>.tgz.sha256)
```

3 Extract the release.

```
tar zxvf nadcm_full-<release>.tgz
```

4 Navigate to the MantaRay HW Manager installation directory `./nadcm-<version>/` and run the upgrade script as a user privileged to run commands with `sudo`.

```
[user@nadcmhost nadcm-<version>]$ sudo ./upgrade -U nuser -P pa55word
```

Note:

You have the following upgrade options:

```
-U, --username      <MANDATORY, user on kubernetes node>
                    the local user on each of the node,
requires sudo
                    permission
-P, --password      <password for the user on kubernetes node>
                    the local user password on each of the node
                    If not provided, the user is expected to
have
                    passwordless sudo on the node
-K, --ssh-key       <path to the SSH private key to login to the
nodes>
                    If not provided, username (-U) and
                    password (-P) are used to login
                    example: -K /home/user/.ssh/node.key
--dont-ask-confirmation <OPTIONAL, don't ask confirmation for
backup>
```

For example:

```
./upgrade -U admin -P 'passw0rd'
```

or:

```
./upgrade -U admin -K /home/user/.ssh/node.key
```

Step result

After running the upgrade command, a confirmation appears to ensure that you have the MantaRay HW Manager backup data. To continue, only yes is accepted.

Result

You have updated the MantaRay HW Manager application.

Postrequisites

After the upgrade, old versions of container images for the MantaRay HW Manager services aren't needed, and Nokia recommends removing them to clean the disk storage space in the MantaRay HW Manager host.

Once the upgrade is successfully finished and MantaRay HW Manager is fully running, you can free the disk storage by running the `prune-images.sh` script:

```
sudo scripts/prune-images.sh
```

The script checks which container images aren't used by the new MantaRay HW Manager release and removes them from the host. It removes only obsolete images that are related to the MantaRay HW Manager application, that is, for services in Kubernetes namespaces: `nadcm` and `monitoring`.

8.1 Upgrading the MantaRay HW Manager on OCP

The procedure for upgrading the MantaRay HW Manager application on OpenShift Container Platform (OCP).

Before you start

See the [Prerequisites for installing the MantaRay HW Manager on OCP](#).

Procedure

- 1 Log in as a non-root user.

Note:

Use the `sudo -l` command to ensure that the non-root user's local account can perform the steps below with `sudo` privileges.

- 2 Prepare the installation package

Download the MantaRay HW Manager installation package to the provisioner/installer node and extract it.

3 Verify prerequisites and configure required environment variables.

Mandatory variables:

- Path to kubeconfig file:

```
# export KUBECONFIG=<path_to_kubeconfig_file>
```

Example:

```
# export KUBECONFIG=/home/admin/kubeconfig_142.yaml
```

- Container manager:

```
# export ContainerManager=podman
```

- Storage class name (get from 'kubectl get storageclass'):

```
# export OCPSTORAGECLASS=<storage_class_name>
```

Example:

```
# export OCPSTORAGECLASS=lvms-vg1
```

- OCP internal or external registry route:

```
# export IMAGEREGISTRY_ADDRESS=default-route-openshift-image
-registry.apps.<cluster_name>.<base_domain>
```

Example:

```
# export IMAGEREGISTRY_ADDRESS=default-route-openshift-image
-registry.apps.cluster01.example.com
```

- Image repository name:

```
# export IMAGEREGISTRY_REPOSITORY=hwm
```

- Cluster admin username with push/pull permissions:

```
# export IMAGEREGISTRY_USERNAME=<cluster-admin username>
```

Example:

```
# export IMAGEREGISTRY_USERNAME=admin
```

- Token from oc login command:

```
# export IMAGEREGISTRY_PASSWORD=<token got from oc login command>
```

- Skip ping test to cluster nodes during installation:

```
# export IGNORE_PING=true
```

Optional variables:

- Temporary folder for HWM images (default: /tmp):

```
# export IMGTMPDIR=/data/hwm/images
```

4 Run upgrade

```
# sudo ./upgrade -U core
```

Expected result:

```
***** BINARY INTEGRITY *****
Note! Execute binary integrity check again using script:
    sudo ./scripts/binary_integrity_check.sh
***WARNING***
Please note that error(s) has been found in binary integrity check.
To setup authentication or to add new users:
    https://nadcmm.apps.mrhwm-sno-2.lab.neat.nsn-rdnet.net/auth/admin/
    username: admin
    password: P@ssw0rd
To login to MantaRay HW Manager:
    https://nadcmm.apps.mrhwm-sno-2.lab.neat.nsn-rdnet.net
    username: nadcmadmin
    password: P@55w0rd
NOTE: Please store all the credentials above carefully
```

9. Removing MantaRay HW Manager and infrastructure

You can remove MantaRay HW Manager with the `sudo ./reset` script, and the infrastructure with the `uninstall-infra.sh` script.

9.1 Removing MantaRay HW Manager

You can remove the MantaRay HW Manager installation with the `sudo ./reset` script.

Purpose

The reset script provided in the software delivery enables you to uninstall the MantaRay HW Manager application. Keep in mind, however, that the script doesn't remove the MantaRay HW Manager infrastructure. To uninstall the infrastructure, you need to use a separate uninstallation script.

Procedure

- 1 To remove the MantaRay HW Manager installation, use the `sudo ./reset` command.

Notice:

The script deletes all MantaRay HW Manager configuration data, including all inventory data.

Notice:

Don't remove the directory structure under `/home/kubedata/install` manually. Directories under this location contain data needed by the infrastructure.

Step result

After running the `sudo ./reset` command, a confirmation appears.

Result

You have removed the MantaRay HW Manager installation.

9.2 Removing infrastructure

You can remove the infrastructure using an uninstall script located in the MantaRay HW Manager installation folder structure.

Purpose

To remove a container runtime manager and Kubernetes, use the `uninstall-infra.sh` script, which is located in the MantaRay HW Manager installation folder structure `./cubeha-infra-*/infra/tools`.

Note:

Uninstalling the infrastructure removes all container images from the local image cache, for the MantaRay HW Manager application and for any other applications that are installed on the same host. MantaRay HW Manager provides all of its images within the installation package, where the images are available for re-installation.

Procedure

- 1 Uninstall Kubernetes and a container runtime manager.

Use the `uninstall-infra.sh` script to uninstall Kubernetes, Helm, and containerd and to perform a cleanup of the nodes.

Step example

```
cd cubeha-infra-0.0.9/infra/tools
sudo ./uninstall-infra.sh -H "10.37.44.138 10.37.44.139 10.37.44.140" -U
nuser -P 'p@ssw0rd'
```

Result

You have uninstalled the infrastructure.

9.3 Uninstalling the MantaRay HW Manager from OCP

The procedure to completely remove the MantaRay HW Manager application from the OpenShift Container Platform (OCP) cluster.

Procedure

- 1 Log in as a non-root user.

Note:

Use the `sudo -l` command to ensure that the non-root user's local account can perform the steps below with `sudo` privileges.

- 2 Set the KUBECONFIG environment variable

```
# export KUBECONFIG=<path_to_kubeconfig_file>
```

Example:

```
# export KUBECONFIG=/home/admin/kubeconfig_142.yaml
```

- 3 Run the reset script

```
# sudo ./reset
```